



Trace element geochemistry of sphalerite in contrasting hydrothermal fluid systems of the Freiberg district, Germany: insights from LA-ICP-MS analysis, near-infrared light microthermometry of sphalerite-hosted fluid inclusions, and sulfur isotope geochemistry

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Abstract

The historic silver mining district of Freiberg (Germany) comprises hydrothermal vein-style mineralization of Permian and Cretaceous age. We compare sphalerite compositions with associated ore-forming fluids and constrain the behavior of critical metals such as In, Ge, and Ga in contrasting hydrothermal environments. Fluid inclusion studies reveal that the Permian veins formed due to boiling and cooling of a low-salinity (0 to 6% eq. $w[\text{NaCl}]$) magmatic-hydrothermal fluid at 350 to 230 °C. In contrast, Cretaceous veins formed by mixing of highly saline (17 to 24% eq. $w[\text{NaCl} + \text{CaCl}_2]$ and variable Na/(Na + Ca) ratios) brines at low temperatures (~120 °C). Sulfides of the Permian ore stage have a narrow range of $\delta^{34}\text{S}_{\text{VCDT}}$ from -2.3 to $+0.9\text{‰}$, while the sulfides of the Cretaceous stage have a large scatter and significantly more negative $\delta^{34}\text{S}_{\text{VCDT}}$ values (-30.9 to -5.5‰), supporting the different nature of the hydrothermal systems. Contrasting fluid systems and ore-forming mechanisms correspond to markedly different trace element systematics in sphalerite. Permian sphalerite is significantly enriched in In (up to 2500 $\mu\text{g/g}$ In) relative to two sphalerite generations of Cretaceous veins. The latter have higher Ge (up to 2700 $\mu\text{g/g}$ Ge) and Ga (up to 1000 $\mu\text{g/g}$ Ga) concentrations. The observed trace element systematics of different sphalerite generations imply that In is enriched in high-temperature, low- to intermediate-salinity fluids with a significant magmatic-hydrothermal fluid component, while Ge and Ga are more concentrated in low-temperature, high-salinity crustal fluids with no obvious magmatic-hydrothermal affiliation.

Keywords Sphalerite · Indium · Germanium · Gallium · Critical metals · EPMA · LA-ICP-MS · Fluid inclusions · Near-infrared light microthermometry · Sulfur isotopes · Geothermometer · Erzgebirge

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Introduction

The silver mining district of Freiberg is well-known for its mining history spanning more than 800 years and for the contribution that research on the polymetallic vein-type mineralization in the Freiberg district has made to the development of mineralogy and economic geology as scientific disciplines (Werner 1791; von Cotta 1850, 1861). Yet, despite extensive research conducted in the past (e.g., Charpentier 1778; Werner 1791; Breithaupt 1849; Müller 1850, 1901; Baumann 1958, 1960), only limited geochemical and fluid inclusion data are available (Thomas 1982; Seifert and Sandmann 2006). Consequently, the mechanisms of ore formation and the origin and nature of the involved fluids are only poorly constrained. The hydrothermal veins are traditionally subdivided according

to their orientation (in general NNE-SSW to N-S and W-E to WNW-ESE) and mineral assemblages. The older N-S striking veins comprise Ag-Pb-Zn-Sb-Cu-Sn-In sulfide mineralization with quartz and carbonate minerals as gangue minerals, while the younger E-W striking veins consist of Ag-Pb-Zn-Ge-Ga sulfide mineralization with fluorite and barite. Recently, the first mineralization stage has been dated to c. 280 Ma (early Permian), and the younger base metal mineralization with fluorite and barite yields a robust Cretaceous age of c. 120 Ma (both are sphalerite Rb-Sr isochron ages, Ostendorf et al. 2018).

Two chemical elements, *indium* (Reich and Richter 1863a, b, 1864) and *germanium* (Winkler 1886), that are regarded as critical or strategic metals (EU Commission 2014), were discovered in base metal ores of the Freiberg mining district in the nineteenth century. Sphalerite is the major host mineral for indium and germanium, as well as gallium in hydrothermal sulfide mineral systems (Bernstein 1985; Möller and Dulski 1993; Höll et al. 2007; Cook et al. 2009, 2011; Frenzel et al. 2014, 2016a, b). The solubility of these elements in ZnS seems to be restricted to very low concentrations (e.g., Patrick et al. 1998 and references therein). However, several studies have shown that the incorporation of tri- and tetravalent cations, like In^{3+} , Ge^{4+} , and Ga^{3+} , into sphalerite is enhanced via coupled substitution with monovalent cations, like Cu^+ and Ag^+ (e.g., Johan 1988; Patrick et al. 1993; Schorr and Wagner 2005; Cook et al. 2012; Belissont et al. 2016). The trace element budget of sphalerite seems to be largely controlled by the origin of the ore-forming fluids in different hydrothermal deposit types (Cook et al. 2009; Belissont et al. 2014), but the temperature of formation also appears to be an important factor (Frenzel et al. 2016a). Particularly, elements such as Fe, Mn, In, and Sn tend to be more concentrated in sphalerite from high-temperature hydrothermal base metal deposits related to magmatic-hydrothermal fluids (e.g., Cook et al. 2009), while other trace elements in solid solution in sphalerite, like Ge, Ga, Tl, and As, are enriched in low-temperature hydrothermal deposits that are not directly related to magmatic-hydrothermal systems (e.g., Belissont et al. 2014).

The hydrothermal vein-type deposit of the historic silver mining district of Freiberg offers the rare chance to study the trace element geochemistry of sphalerite in different appearances (coarse-grained and colloform sphalerite) and hydrothermal environments that significantly differ with respect to their temperature, salinity, acidity, and ore-forming process (i.e., cooling, boiling, or mixing). Here, the combined characterization of trace elements in sphalerite (EPMA and LA-ICP-MS analysis) and their associated fluids (microthermometry and isotopic composition of sulfides) is carried out to evaluate the mobility, source, and deposition of critical metals in hydrothermal solutions. The investigation of trace element systematics in sphalerite of the different well-characterized hydrothermal vein types is furthermore an ideal opportunity to learn

more about In, Ge, and Ga as petrogenetic proxies for ore deposition and fluid evolution (cf. Cook et al. 2009; Belissont et al. 2014; Frenzel et al. 2016a).

Regional geology

The polymetallic hydrothermal veins of the Freiberg ore district are hosted by gneisses and mica schists of the Variscan basement of the Erzgebirge. The Variscan orogenic belt formed as a consequence of the collision of Gondwana and Laurussia (e.g., Matte 1991; Franke 2000; Nance et al. 2010; Kroner and Romer 2013). The Erzgebirge is part of the Saxo-Thuringian zone of the Variscides and is located at the northern margin of the Bohemian Massif (Kossmat 1927). The peak metamorphic conditions occurred at about 340 Ma (Tikhomirova et al. 1995; Kröner and Willner 1998; von Quadt and Günther 1999; Werner and Lippolt 2000). All metamorphic units of the deeper basement were exhumed to upper crustal levels at about 330 Ma (Kroner et al. 2010; Kroner and Götz 2010).

In the late Carboniferous, from about 327 to 300 Ma, several geochemically different late-collisional Variscan granites, microgranites, and subvolcanic dikes, as well as distinct groups of lamprophyres, intruded into the metamorphic rocks (Förster et al. 1999; von Seckendorff et al. 2004; Seifert 2008; Förster and Romer 2010). The onset of Permian rifting (breakup of Pangea) and related magmatism (c. 300 to 275 Ma) is recorded by subvolcanic dikes and minor granitic intrusions (e.g., Nasdala et al. 1998; Kempe et al. 2004; Förster and Rhede 2006; Förster et al. 2007), as well as rhyolites and pyroclastics in Permian sedimentary basins (Schneider and Romer 2010; Hoffmann et al. 2013; Repstock et al. 2017). Spatially and temporally associated with the late to post Variscan felsic magmatism are W(-Mo) and Sn-W(-Li) greisen, skarn, and polymetallic vein deposits (e.g., Seifert and Kempe 1994; Romer et al. 2007; Breiter 2012; Seifert and Pavlova 2016; Ackerman et al. 2017; Zhang et al. 2017). Throughout the early Mesozoic, the northern Erzgebirge was probably at the margin of the Central European Basin (Feist-Burkhardt et al. 2008). Fluvial sediments of Upper Cretaceous age were deposited in paleorivers extending toward the NE into the Saxonian Cretaceous Basin and are preserved in depressions of the Western Erzgebirge (Voigt 2009; Janetschke and Wilmsen 2014). The northern parts of the Bohemian Massif, and with it the Erzgebirge, have experienced uplift and denudation since the Oligocene or early Miocene (Ziegler and Dèzes 2007).

Local geology

The central Freiberg district is situated at the center of an ellipsoid-shaped gneiss structure that is enveloped by

mica schists and phyllites. Many other peripheral mining districts (e.g., Bräunsdorf, Kleinvoigtsberg, Tharandt) occur in the outer zones of this ellipsoid-shaped gneiss structure (Baumann et al. 2000; Burisch et al. 2018a). The gray gneisses can be subdivided into two units that are different with respect to their fabric and mineral composition. The core is composed of medium- to coarse-grained biotite-plagioclase(oligoclase)-orthoclase gneisses (Müller 1901) that are interpreted to derive from Cadomian (c. 540 Ma) granodiorite protoliths (Kröner et al. 1995; Tichomirowa et al. 2001, 2012). The surrounding rock units are fine- to medium-grained biotite-muscovite-plagioclase gneisses that probably originate from greywackes (Tichomirowa et al. 2012; Fig. 1). The foliation of both rock units dips around 30° radially away from the center indicating a dome-like geometry. A late-collisional granite (c. 325 to 320 Ma, Tichomirowa 1997)

intruded into the eastern part of the Freiberg gneiss dome. Igneous rocks of the Tharandt Volcanic Complex (also presumably c. 320 Ma, Sebastian 2013 and references therein) are located to the east of the Freiberg gneiss dome (Fig. 1). Carboniferous porphyritic microgranite dikes are likely related to the magmatism in the Teplice-Altenberg and the Tharandt Volcanic Complexes (Hoffmann et al. 2013). The youngest generation of lamprophyres (c. 300 Ma) represents an upper age limit for cross-cutting Permian hydrothermal veins (cf. Seifert 2008). The dykes exploited faults in the gneisses that occasionally acted also as pathways for Permian and Cretaceous hydrothermal fluids (Fig. 1). The Permian mineralization is typically present on N-S and NNE-SSW trending brittle fault structures that are related to the postorogenic Permian extension. Younger Cretaceous mineralization occurs dominantly on E-W trending faults

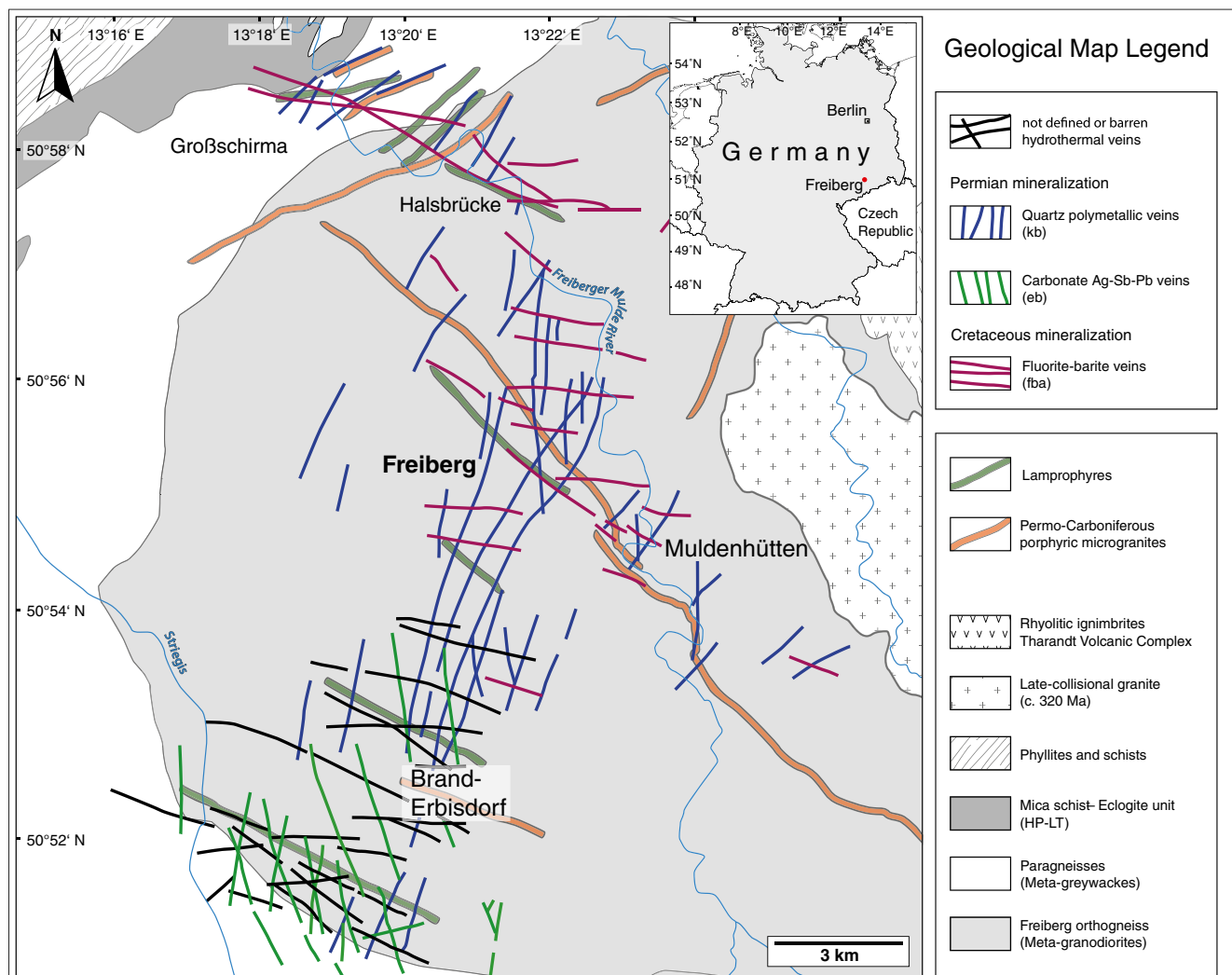


Fig. 1 Schematic geological map of the historic silver mining district of Freiberg with major hydrothermal vein generations. Modified after Müller (1901), Baumann et al. (2000), and locations of lamprophyres after Seifert (2008)

that opened contemporaneously with the opening of the northern Atlantic (Baumann et al. 2000; Ostendorf et al. 2018); however, due to reactivation of Permian structures, younger mineralization may also occur in N-S and NNE-SSW trending faults.

Hydrothermal vein types

The hydrothermal veins of the Freiberg district formed during two major ore-forming periods: the first is of early Permian (c. 280 Ma) and the other of Cretaceous (c. 120 Ma) age (Ostendorf et al. 2018). The so-called five-element (Ag, Bi, Co, Ni, As) veins characteristic elsewhere in the Erzgebirge do play only a very minor role in the Freiberg district (Müller 1901; Baumann 1958, 1960).

The early Permian mineralization event is represented by a polymetallic-sulfide-quartz paragenesis (“kiesigblendige Bleierzformation”, abbrev. *kb*) and a Ag-Sb-carbonate-quartz paragenesis (“edle Braunsparformation”, abbrev. *eb*) (Baumann et al. 2000; Seifert and Sandmann 2006). The *kb* veins comprise arsenopyrite, pyrite, Fe-rich sphalerite, stannite, cassiterite, chalcopyrite, galena, minor Ag-sulfides, and the gangue minerals quartz, minor chlorite, and carbonates, like dolomite and calcite (Figs. 2 and 3a). The first generation of quartz (Qtz I) in the *kb* veins forms euhedral, transparent comb-like crystals and commonly appears at the selvages of the hydrothermal vein infill (Müller 1901). A second generation of quartz (Qtz II), which formed coevally with the sulfide minerals (sphalerite, stannite, and chalcopyrite), forms also euhedral crystals (cf. Fig. 7a, d). These crystals characteristically exhibit milky white zones of microcrystalline quartz at the rims, while they are clear in the center. Quartz II is the dominant gangue mineral in the *kb* veins (Müller 1901). Chlorite is very common in the *kb* veins and forms clusters in sulfide ore or is trapped on crystal planes in euhedral comb quartz (cf. Fig. 7). Ivory-colored milky carbonates overgrow Qtz II in cavities or cross cut the *kb* veins as veinlets.

The *eb* veins comprise, in addition to galena, sphalerite, pyrite, arsenopyrite, freibergite, and chalcopyrite, various kinds of Ag-, Sb-, and Ag-Sb-As-sulfides, such as acanthite, berthierite, boulangerite, stibnite, freieslebenite, miargyrite, pyrrargyrite, proustite, or stephanite (Figs. 2 and 3g). Characteristic gangue minerals in the *eb* veins are quartz and many different types of carbonates (Fig. 2). The carbonate mineralogy seems to change from early Fe- and Mn-rich varieties like siderite and rhodochrosite to ankerite, Fe-bearing dolomite, and late dolomite and calcite in cavities and veinlets (Müller 1901).

The Cretaceous hydrothermal veins are rich in fluorite and barite, giving rise to the name “Fluorbarytische

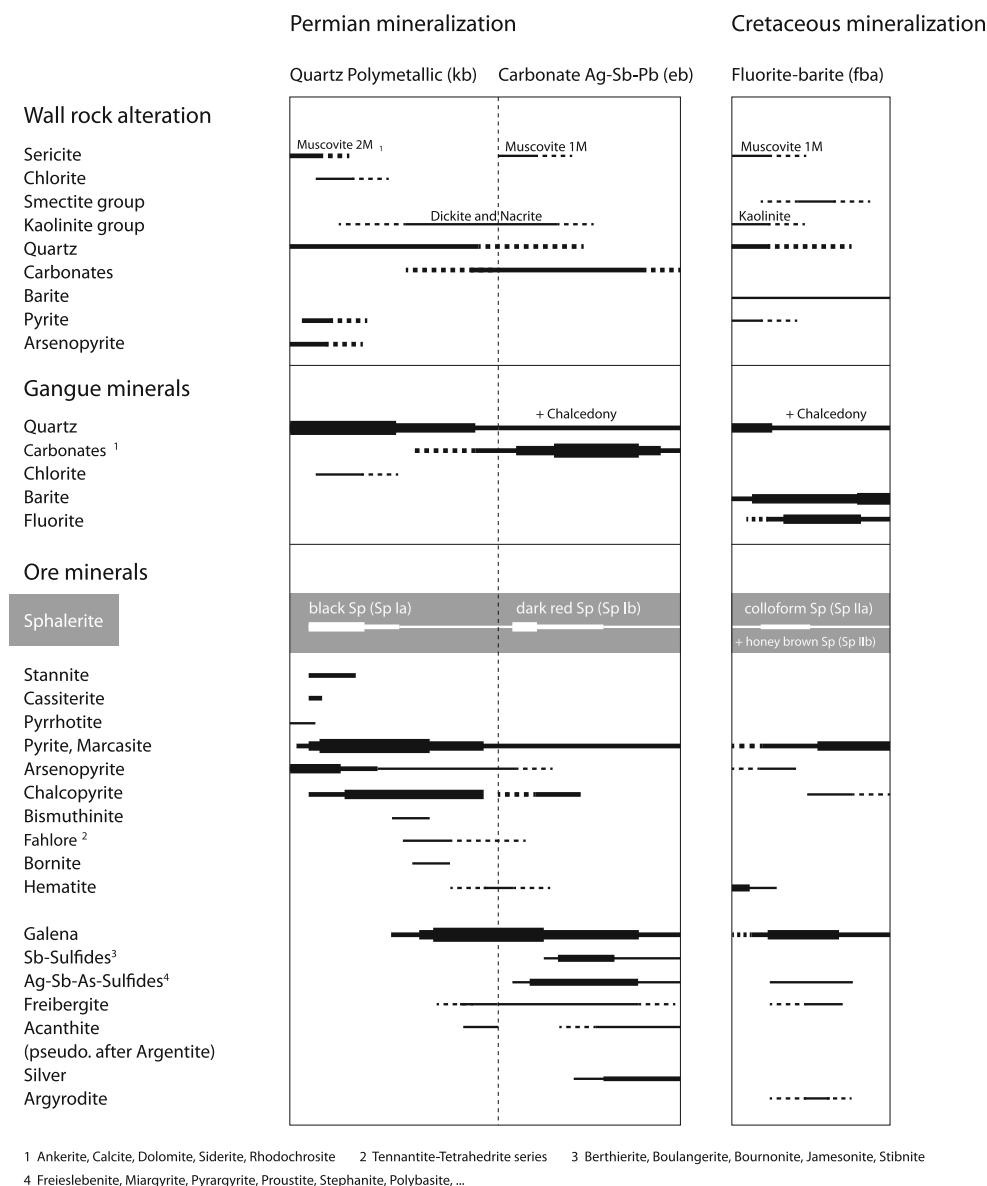
Bleierzformation” in German (or *fba* for short). Quartz and chalcedony are also found as gangue minerals in the *fba* veins. Colloform and coarse-grained sphalerite, galena, pyrite/marcasite, arsenopyrite, and Ag-minerals are common ore minerals in the *fba* veins. Gangue minerals and sulfides form thin rhythmic bands (a few millimeters to centimeters wide) in the *fba* veins. Pseudomorphous replacement of barite and fluorite by microcrystalline quartz and chalcedony is common in the *fba* veins and is most likely related to cooling after ore formation (Burisch et al. 2017). A simplified paragenetic diagram (Fig. 2) illustrates the different minerals associated with *kb*, *eb*, and *fba* veins.

Intensive wall rock alteration associated with the hydrothermal vein type mineralization in the Freiberg district varies significantly from less than a few centimeters to about 0.5 m width, rarely more than 2 m (Rösler and Kühne 1970). Besides disseminated quartz and sulfides (arsenopyrite and pyrite) of the Permian *kb* and *eb* veins, chlorite, muscovite (2M₁), epidote, and kaolinite group minerals (here dickite and nacrite polymorphs) occur in the alteration zones (Fig. 2). In contrast, wall rock alteration zones related to the Cretaceous *fba* type mineralization are usually less well pronounced and comprise “sericite” (muscovite 1M), mixed-layer illite-smectite minerals, and kaolinite (Rösler and Kühne 1970).

Spatial distribution of vein types across the Freiberg district

The three most important hydrothermal mineralization types (Permian *kb* and *eb*, Cretaceous *fba*) are developed in veins with distinct regional distribution and distinctive major strike directions across the Freiberg district. The *kb* type dominates the central district around Freiberg in veins that strike mainly N-S and NNE-SSW (Fig. 1). The *eb* type is particularly well-developed on NNE-SSW and NNW-SSE striking veins in the southern district around the town of Brand-Erbisdorf. It also occurs along the western and northwestern margins of the district resulting in a distinct lateral zonation. The presumed close genetic link between the younger *eb* veins and the older *kb* veins based on field observations has recently been confirmed by Rb-Sr and Pb-Pb isotope systematics (Ostendorf et al. 2018). In contrast, the Cretaceous *fba* mineralization is predominantly found on E-W and WNW-ESE striking hydrothermal veins in the northern and central portions of the district (Fig. 1). South of Freiberg, the predominantly E-W striking hydrothermal veins are significantly less mineralized (i.e., little sulfides, but quartz, barite, and clay minerals) and were mapped as barren veins in the past (Müller 1901).

Fig. 2 Simplified paragenetic mineral sequences of the hydrothermal veins of the historic Freiberg ore district, showing the Permian and Cretaceous mineralization. Note that the younger Ag-Bi-Co-Ni-As mineralization was omitted for perspicuity. Dotted lines indicate uncertain occurrence and line thickness roughly illustrates the relative mineral abundance. The appearances of different sphalerite species are highlighted. Based on our observations and reported data from Müller (1901), Baumann (1958, 1994), and Baumann et al. (2000). Wall rock alteration constructed from XRD and DTA results by Rösler and Kühne (1970)



Materials and methods

Sampling and sample preparation

The sample suite comprises ore specimens from the historic mineral and ore deposit collection of the TU Bergakademie Freiberg and recently collected ore samples from the Himmelfahrt (Reiche Zeche shaft) underground mine (Table ESM 1). Most of the kb vein samples are from the central district in Freiberg, while all of the eb vein samples are from the southern district around Brand-Erbisdorf (Table ESM 1). Samples of the Cretaceous fba veins are both from the northern (Halsbrücke) and central district of Freiberg. Detailed information on the sample localities is given in Table ESM 1. In total, 69 samples were selected for this study that contained sufficiently

large grains of sphalerite to conduct trace element analyses and microthermometry on the same sample, and additionally contain gangue minerals that are paragenetically related to sphalerite. Many of the samples were also used for sphalerite Rb-Sr geochronology (Ostendorf et al. 2018).

Thin sections (30 µm) and polished sphalerite grain mounts were prepared for ore microscopy, scanning electron microscopy (SEM), electron probe microanalysis (EPMA), and laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS). Sphalerite, galena, pyrite, and arsenopyrite for sulfur isotope analysis were obtained with a microdrill and then optically screened for purity under a stereomicroscope. Double polished thin sections (70 to 250 µm thick) for fluid inclusion studies were made from a selection of representative samples (Table ESM 1).

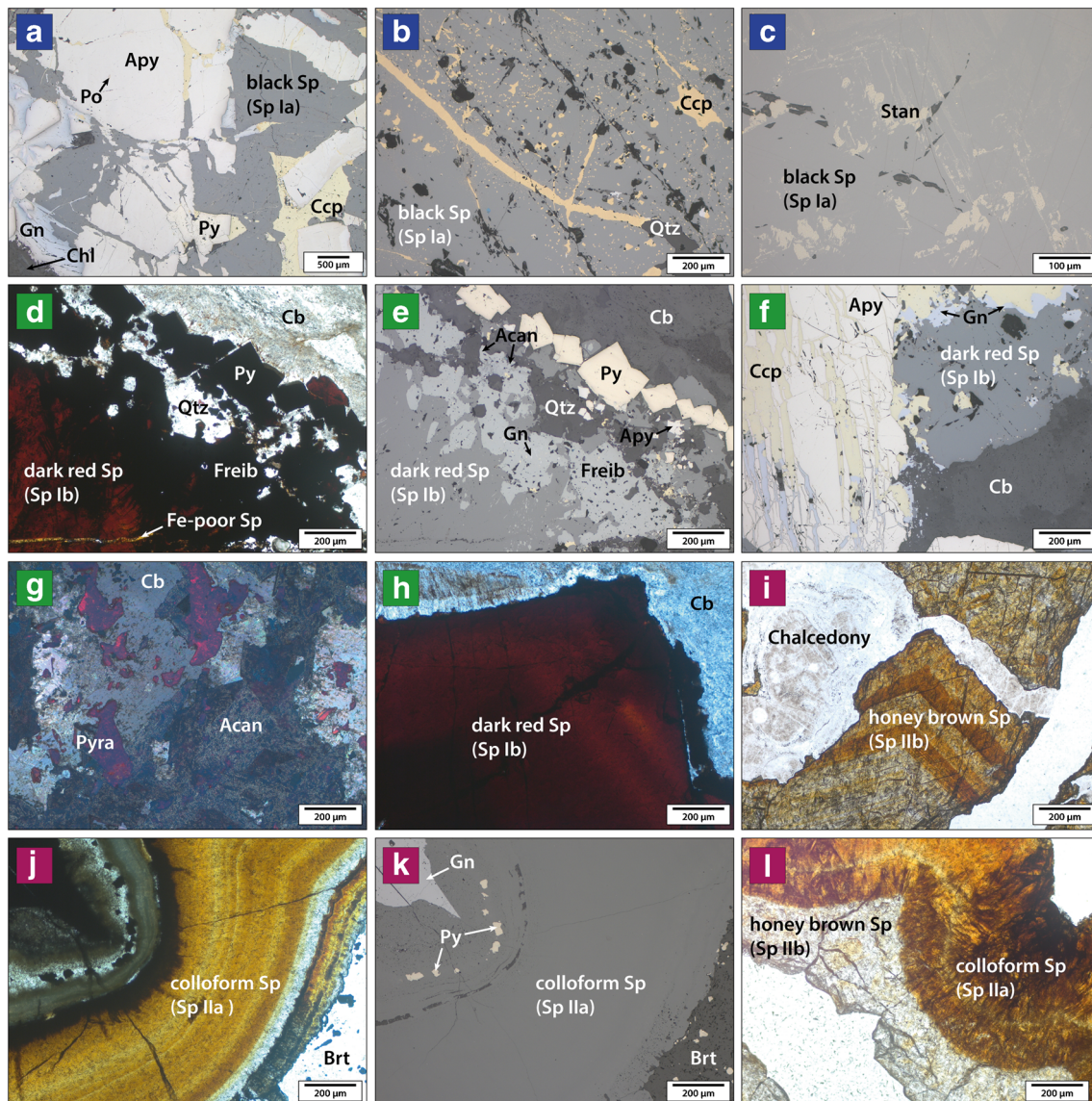


Fig. 3 Photomicrographs of sphalerite (Sp) and representative ore mineral assemblages from the Freiberg hydrothermal veins. **a** Sphalerite Ia from the kb veins, chalcopyrite (Ccp), galena (Gn), and chlorite (Chl) replacing white arsenopyrite (Apy), cream-white pyrite (Py), and brownish cream-white pyrrhotite (Po); reflected light. **b** Sphalerite Ia with fine-grained DIS of chalcopyrite and the gangue mineral quartz (Qtz); reflected light. **c** Sphalerite Ia and olive-gray stannite (Stan) highlighting growth zoning in sphalerite; reflected light. **d, e** Sphalerite Ib (reddish translucent) from eb veins with Fe-poor sphalerite (brownish translucent), galena, pyrite, arsenopyrite, greenish gray freibergite (Freib), brighter gray acanthite (Acan), quartz, and carbonates (Cb); transmitted

and reflected light, respectively. **f** Sphalerite Ib, galena, chalcopyrite, and carbonates replacing arsenopyrite; reflected light. **g** Red internal reflections of pyrrhotite (Pyra) and dark acanthite in carbonates; cross-polarized reflected light. **h** Reddish translucent sphalerite Ib in carbonates; cross-polarized transmitted light. **i** Zoned coarse-grained sphalerite IIb (honey brown) with rhythmic banding and chalcedony from the fba veins; transmitted light. **j, k** Colloform sphalerite IIa from the fba veins with colorless cleiophane layers, galena, pyrite, and barite (Brt); transmitted and reflected light. **l** Colloform sphalerite IIa overgrown by coarse-grained honey brown sphalerite IIb; transmitted light

SEM and EPMA analyses

Analyses were carried out using SEM-based image analysis (Mineral Liberation Analyzer, MLA) and EPMA. An FEI Quanta 650F SEM-MLA system equipped with two Bruker Quantax X-Flash 5030 energy-dispersive X-ray spectrometers (with software version 3.1.4) at the

Institute of Mineralogy of the Technische Universität Bergakademie Freiberg was used for automated data acquisition (Fandrich et al. 2007). Thin sections of different ore samples containing sphalerite and polished grain mounts with hand-picked sphalerite grains were mapped in the grain-based X-ray mapping (GXMAP) mode (Bachmann et al. 2017). This technique was applied to

identify sphalerite samples that are free of visible chalcopyrite, stannite, or any other unwanted solid inclusions that could potentially distort the trace element analyses of sphalerite (EPMA and LA-ICP-MS analyses). Further analytical details are given in Table ESM 2.

EPMA measurements were conducted at the Helmholtz Institute Freiberg for Resource Technology (HIF) using a JEOL JXA-8530F equipped with a field emission gun (FEG) and five wavelength-dispersive spectrometers (WDS). Analyses were carried out at an accelerating voltage of 20 kV and a beam current of 30 nA, and a defocused beam of 15 μm diameter was used in order to reflect the spot sizes of subsequent LA-ICP-MS analyses on the same sample spot. Counting times, peak positions, measurement times on backgrounds, detection limits, and further analytical conditions are given in Table ESM 3. Interferences of the Cd-L β 1 and Sn-LN lines on the In-L α line were corrected using the offline correction approach described in Osbahr et al. (2015). Settings for the EPMA mappings are also included in the Electronic supplementary material (Table ESM 3).

Laser ablation inductively coupled plasma mass spectrometry

Trace element analyses of sphalerite were performed using a LA-ICP-MS system at the Friedrich-Alexander-Universität Erlangen-Nürnberg, consisting of a 193-nm New Wave Research Excimer laser and an Agilent 7500i quadrupole ICP-MS. The ICP-MS was operated with a plasma power of 1320 W, and He (0.65 L/min) and Ar (1.09 L/min) were chosen as carrier gases. Additionally, Ar was used as plasma (14.9 L/min) and auxiliary gas (0.9 L/min). The instrument was tuned to low oxide rates of $< 0.4\%$ ThO $^+$ /Th $^+$. The laser ablation system was operated with a repetition rate of 20 Hz, a beam spot size of 25 μm , an irradiance of 0.74 GW/cm 2 , and a fluence of 3.7 J/cm 2 . The following isotopes were measured each for 10 ms at the peak maximum: ^{29}Si , ^{33}S , ^{34}S , ^{57}Fe , ^{66}Zn , and ^{68}Zn . All other isotopes were measured each for 25 ms at the peak maximum: ^{55}Mn , ^{59}Co , ^{60}Ni , ^{61}Ni , ^{63}Cu , ^{69}Ga , ^{73}Ge , ^{75}As , ^{77}Se , ^{95}Mo , ^{109}Ag , ^{111}Cd , ^{115}In , ^{118}Sn , ^{119}Sn , ^{121}Sb , ^{202}Hg , ^{205}Tl , ^{208}Pb , and ^{209}Bi . This results in a time span of 0.6092 s for each mass scan. Each LA-ICP-MS analysis comprises 20 s of background measurement and 20 s of sample ablation (signal measurement), followed by a 60-s retention time to ensure a proper cell washout. The relative standard deviation (RSD) for analyses ranged from 2 to 12% (dependent on the concentration) and averaged 5%. An exception represents ^{73}Ge and ^{205}Tl that mostly yielded values greater than 10% RSD, since Ge and Tl concentrations in sphalerite Ia and Ib ranged close to the respective minimum limits of detection (LODs). Trace

element concentrations and LOD for each analysis were calculated using the algorithms presented in Longerich et al. (1996). Time-resolved LA-ICP-MS signals that showed clear peaks or strong rises (possible mineral inclusions) were rejected, and only smooth ablation signals were integrated to a measurement interval. To calculate the absolute concentration from the uncorrected LA-ICP-MS count rate, Zn values of each corresponding EPMA analysis were used as internal standard. External calibration was performed with the georeference material MASS-1 (Wilson et al. 2002). Values for trace element concentrations were taken from an updated and revised preliminary certificate of analysis from 2015 (Wilson, personal communication). The synthetic standard glasses NIST SRM 610 and 612 (Pearce et al. 1997) were additionally measured to recognize a possible instrumental drift. Furthermore, the synthetic polymetallic sulfide material MASS-1 was used as external Ge standard, having a reported concentration of 57.8 ± 2.6 $\mu\text{g/g}$ Ge (cf. Belissont et al. 2014). Indium concentrations were corrected for the isobaric interference of ^{115}In and ^{115}Sn . The elements Ni, Mo, and Bi are in all analyses below the limits of detection. The complete LA-ICP-MS dataset is included in the Electronic supplementary material (Table ESM 4). A linear correlation between the geometric sample means (Fe, Mn, and In) of EPMA and LA-ICP-MS analyses of the sphalerite samples (Table ESM 5) shows that the data of the two methods fit very well (R^2 (Fe) = 0.98; R^2 (Mn) = 0.98; and R^2 (In) = 0.94). A direct comparison (EPMA vs. LA-ICP-MS) for all other elements is not possible as the majority of trace elements in sphalerite are below the limit of quantification (LOQ) of the EPMA method.

Principal component analysis

Principal component analysis (PCA) is a multivariate statistical exploratory technique to project and interpret multivariate datasets. Since a PCA does not properly work with incomplete datasets (Van den Boogaart and Tolosana-Delgado 2013), values below the LOD in the LA-ICP-MS dataset were therefore replaced by random imputation with a normal distribution of the mean and standard deviation equal to the respective detection limit (cf. Frenzel et al. 2016a). In particular, the values of Ge (40% < LOD), Tl (39%), As (17%), In (14%), Sb (13%), and Co (7%) are below the limits of detection in certain sphalerite species. Selenium was omitted completely from the dataset and imputation because more than 54% of all Se values are below the limit of detection. Negative values that resulted from the imputation procedure were transformed into absolute values. The raw concentrations of the manipulated sphalerite trace element dataset were further log-transformed for the PCA.

Sulfur isotope geochemistry

Sulfur isotope analysis of metal sulfide separates was carried out at the Geologisch-Paläontologisches Institut, Westfälische Wilhelms-Universität Münster. Isotope measurements were performed using a Carlo Erba elemental analyzer interfaced to a ThermoFinnigan Delta Plus isotope ratio mass spectrometer (EA-IRMS). Reference materials for sulfur isotope measurements were NBS-127, IAEA-S-1, IAEA-S-2, IAEA-S-3, and an in-house Ag_2S reference material. Repeatability was determined by replicate analyses of the reference materials and was better than $\pm 0.3\text{‰}$ (1σ). Results of sulfur isotope analysis and further analytical details are given in the Electronic supplementary material (Table ESM 6).

Fluid inclusion microthermometry

Fluid inclusion petrography and the definition of *fluid inclusion assemblages* (FIAs) in their relative chronology were done according to Diamond (1990) and Goldstein and Reynolds (1994). Fluid inclusions were classified as primary (p), pseudosecondary (ps), secondary (s), or isolated inclusions (iso) with no clear spatial relation to growth zones or fractures (cf. Walter et al. 2015). Fluid inclusion studies in quartz, fluorite, and sphalerite were carried out on double polished thick sections with variable thickness of 70 to 250 μm (depending on the transparency of sphalerite). Microthermometry was carried out using an Olympus BX53 microscope equipped with a Linkham THMSG600 heating-freezing stage. Microthermometric measurements in opaque sphalerite were done using a QImaging Retiga 2000R CCD digital camera that is sensitive to near-infrared light up to a wavelength (λ) of 1000 nm. To prevent heating of the silver block by infrared radiation, the field and aperture diaphragm were closed as much as possible and the light intensity was reduced to the lowest practicable setting (cf. Moritz 2006; Lüders 2017). Calibration of the heating and freezing stage was done daily monitored by analyzing synthetic H_2O - CO_2 and pure H_2O fluid inclusion standards before and after each measurement campaign (CO_2 melting at -56.6 °C , melting of H_2O at 0.0 °C and the homogenization temperature of pure H_2O at 374 °C). Phase changes were measured with a temperature rate of 1 °C/min . Resulting uncertainties are $\pm 0.2\text{ °C}$ for melting temperatures and $\pm 1\text{ °C}$ for homogenization temperatures. Melting temperatures of hydrohalite, T_{m} (hh), and ice, T_{m} (ice), of each individual fluid inclusion were measured in three runs and the results are given as average values of these triplets. Liquid and vapor volume fractions (φ_{liq} and φ_{vap}) for each fluid inclusion were estimated with volume fraction tables of Shepherd et al. (1985) at 25 °C . The

salinities, % eq. $w(\text{NaCl} + \text{CaCl}_2)$ and % eq. $w(\text{NaCl})$, of individual fluid inclusions in the ternary H_2O - NaCl - CaCl_2 system were calculated according to Steele-MacInnis et al. (2011) and in the binary NaCl - H_2O system with the Excel spreadsheet of Steele-MacInnis et al. (2012), respectively. Fluid inclusions showing signs of postentrapment processes (e.g., necking down) or indications of leaking due to microfractures were excluded from the dataset (Table ESM 7).

Laser Raman analyses of fluid inclusions were conducted on a confocal Horiba Jobin Yvon LabRAM HR instrument at the Helmholtz Institute Freiberg for Resource Technology using an 485-nm laser, an $\times 100$ objective ($\text{NA} = 0.9$), a slit width of 100 μm , and a confocal hole of 500 μm . The Raman spectrometer was calibrated using the 520.7-cm^{-1} Raman band of a Si wafer. The acquisitions were performed using gratings of 600 and 1800 lines/mm, at variable integration times and acquisition runs to record liquid and vapor phase as well as host mineral spectra. Raman spectra for the detection of CO_2 were obtained at 120 s counting time with three acquisitions. The limit of detection for this setting is better than $0.13\text{ mol\% CO}_2$ (Rosso and Bodnar 1995). Raman spectra are presented in a range from 200 to 4000 cm^{-1} .

Results

Petrographic observations

Based on optical characteristics, several sphalerite generations were distinguished. *Sphalerite Ia* from the kb veins is an opaque Fe-rich sphalerite variety (*christophite* or *marmatite*) with a black appearance in hand specimen. It is commonly subhedral and intergrown with associated quartz, chalcopryrite, arsenopyrite, pyrite, galena, and stannite (Fig. 3a). Small inclusions of chalcopryrite formed by diffusion-induced segregation (DIS) are present in sphalerite Ia (Fig. 3b). Stannite inclusions in growth zones are common (Fig. 3c), and occasionally, small quantities of cassiterite and pyrrhotite may occur in massive sphalerite Ia. *Sphalerite Ib* from the eb veins is an Fe-rich sphalerite that appears opaque to dark red in transmitted light (Fig. 3d, h). It is associated with Ag- and Ag-Sb-As-sulfides (e.g., acanthite, freibergite, pyrargyrite) and galena (Fig. 3e, g). In contrast to sphalerite Ia, sphalerite Ib commonly lacks inclusions of chalcopryrite and stannite (Figs. 2 and 3e, f).

Sphalerite from the Cretaceous fba veins occurs as colloform or coarse-grained sphalerite species. Colloform *sphalerite IIa* shows typical color (dark brown to pale white) and textural variations between discrete layers. The core of the colloform sphalerite is commonly formed

Table 1 EPMA and LA-ICP-MS results for the different sphalerite species. Sphalerite Ia from kb veins, sphalerite Ib from the eb veins, and colloform sphalerite IIa and coarse-grained honey brown sphalerite IIb from the fba veins. Zn and Fe are given in weight percentage (EPMA), and all other trace elements in micrograms per gram (LA-ICP-MS). Averages are presented as geometric mean values of all sample geometric means. Minimum (min) and maximum (max) concentrations are extreme values of single sample geometric means

		Zn	Fe	Mn	Cd	Cu	Ag	In	Ga	Ge	Sn	Co	As	Sb	Tl	Pb	Hg	Se
		(wt%)	(wt%)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)	(μg/g)
Sphalerite Ia	Geom. mean	52.3	11.9	2079	5725	1266	14	300	4.9	0.4	46	5.6	0.2	0.3	<0.1	1.0	7.7	2.0
	Min	50.2	5.3	428	3809	402	6.1	14	0.6	<LOD	3.4	0.1	<LOD	<LOD	<LOD	0.1	2.5	0.8
	Max	57.1	13.9	12,392	7472	4012	34	1226	87	26	1290	132	0.9	3.1	0.5	5.7	31	11
Sphalerite Ib	Geom. mean	54.5	8.2	2187	2689	538	25	9.3	7.5	0.4	81	0.7	0.5	3.2	<0.1	2.9	12	1.0
	Min	52.0	2.0	359	1176	55	3.6	0.6	1.6	<LOD	4.0	<LOD	<LOD	<LOD	<LOD	0.3	5.1	0.6
	Max	59.0	12.9	5972	6344	5081	214	360	24	0.6	825	48	1.4	23	0.2	40	35	2.1
Sphalerite IIa	Geom. mean	63.4	1.4	204	656	32	163	0.8	32	48	0.8	0.1	88	3772	769	7965	19	1.8
	Min	61.8	0.7	148	128	<LOD	44	<LOD	9.2	19	<LOD	<LOD	17	243	105	2785	5.9	<LOD
	Max	64.8	2.0	420	2707	2153	1073	45	53	1716	4.0	0.4	1992	11,480	2732	16,754	65	2.3
Sphalerite IIb	Geom. mean	65.6	0.2	2.2	3381	331	42	3.2	91	3.3	8.9	5.1	6.1	220	0.6	66	23	2.0
	Min	64.7	0.1	<LOD	1356	40	1.8	0.4	7	<LOD	0.9	0.8	<LOD	8.0	<LOD	2.1	6.7	<LOD
	Max	66.2	0.3	13	5868	3242	458	27	519	14	58	15	73.1	4350	20	623	492	3.4

by galena. Pyrite grains often accumulate in layers of pale white, Fe-poor sphalerite, termed *cleiophane* (Fig. 3j, k). Coarse-grained *sphalerite IIb* exhibits typical growth and sector zoning and has a honey brown appearance (Fig. 3i). Colloform sphalerite IIa and coarse-grained sphalerite IIb are intimately intergrown with fluorite, barite, quartz, and chalcedony. Occasionally, colloform sphalerite IIa is overgrown by later coarse-grained honey brown sphalerite IIb (Fig. 3l).

EPMA and LA-ICP-MS data of sphalerite

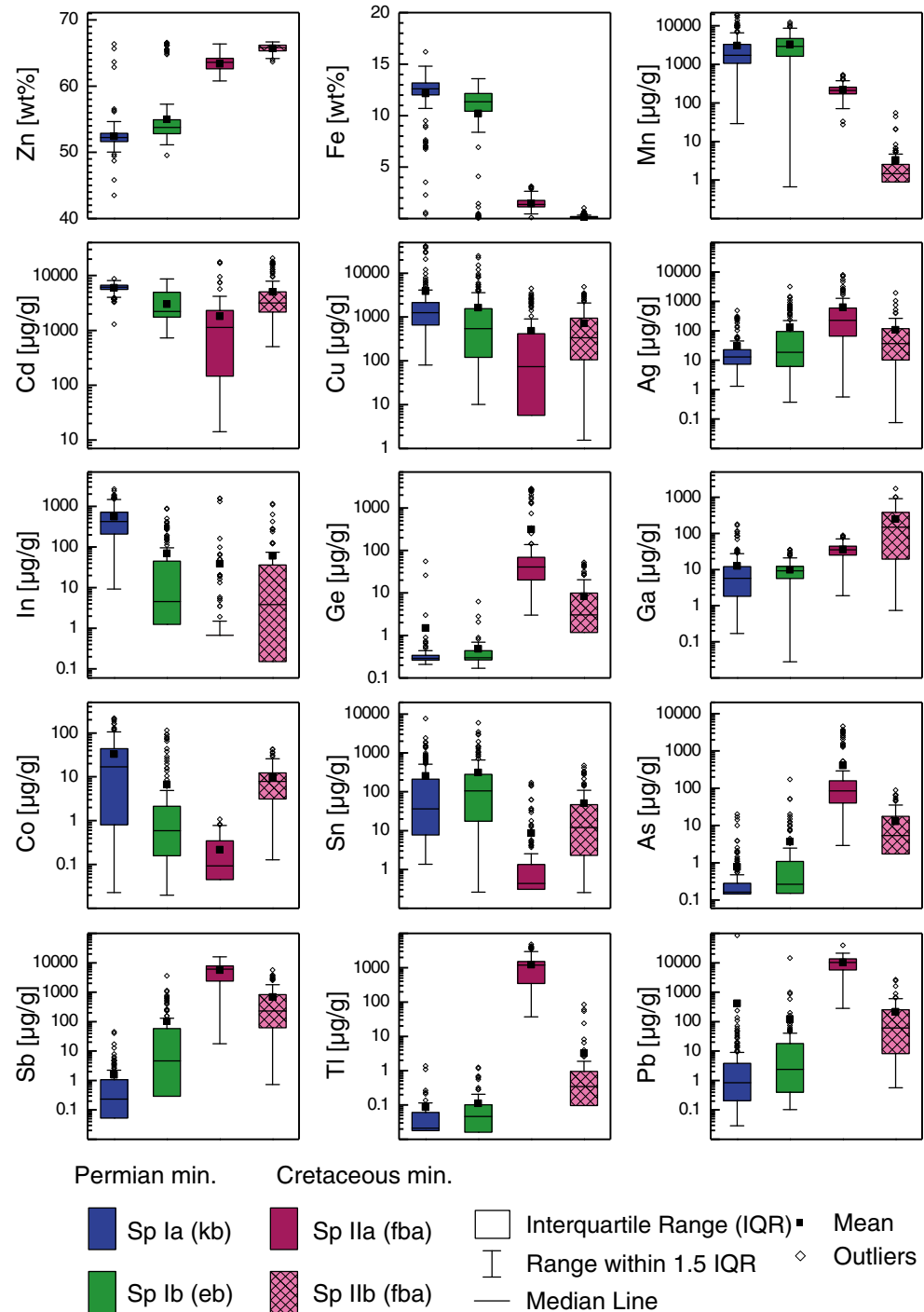
Major (Zn and Fe from EPMA) and trace element compositions (LA-ICP-MS) of the four textural sphalerite varieties are presented in Table 1. The elemental averages of the sphalerite types were calculated hierarchically from geometric sample means. The entire trace element dataset includes 205 analyses of sphalerite Ia (26 samples), 156 analyses of sphalerite Ib (16 samples), 135 analyses of sphalerite IIa (11 samples), and 100 analyses of sphalerite IIb (10 samples), adding up to a total of 596 EPMA and corresponding LA-ICP-MS analyses (Table ESM 4). EPMA and LA-ICP-MS analyses of the four sphalerite types are additionally presented in compiled box-and-whisker plots (Fig. 4) to provide an overview of the data including median, arithmetic mean, possible outliers, and variance of the element concentrations.

Sphalerite Ia of the Permian kb veins has generally higher Fe (11.9 wt%) and Cd (0.6 wt%) concentrations than sphalerite Ib (8.2 wt% Fe and 0.3 wt% Cd; Table 1) of the Permian eb veins. The trace elements In, Cu, and Co are significantly enriched in sphalerite Ia (300 μg/g In, 1270 μg/g Cu, and 6 μg/g Co) relative to sphalerite Ib (9 μg/g In, 540 μg/g Cu, and 0.7 μg/g Co; Table 1). Indium mean concentrations reach up to 1200 μg/g in some samples of sphalerite Ia (kb veins). Manganese concentrations in both Permian sphalerite types are more or less similar within the uncertainties (2080 μg/g Mn for Sp Ia and 2190 μg/g Mn for Sp Ib). Mean concentrations of Sn tend to be slightly higher in Sp Ib in comparison to Sp Ia (50 μg/g Sn for Sp Ia and 80 μg/g for Sp Ib; Table 1 and Fig. 4). Concentrations of Ge and Ga in both Permian sphalerite types (Sp Ia and Sp Ib) are low (< 5 μg/g) or close to the respective LODs (Table 1 and Fig. 4).

In contrast to the Permian sphalerites Ia and Ib, Cretaceous sphalerites IIa and IIb from the fba veins exhibit very low Fe (1.4 wt% Fe for Sp IIa and 0.2 wt% Fe for Sp IIb) and Mn (200 μg/g Mn for Sp IIa and 2 μg/g Mn for Sp IIb) concentrations, while Ge, Ga, Pb, As, Tl, and Sb are notably elevated (cf. Fig. 4). Colloform sphalerite IIa has remarkably high concentrations of Ge (geometric mean of 50 μg/g Ge, up to about 1700 μg/g Ge), Pb (8000 μg/g), As (88 μg/g), Tl (800 μg/g), and Sb (3800 μg/g; Table 1). Hg concentrations are elevated in coarse-grained sphalerite IIb (20 μg/g Hg, up to 500 μg/g

Hg). While coarse-grained sphalerite IIb of the fba veins shows very low Fe (0.2 wt%) and Mn concentrations (2 $\mu\text{g/g}$), Ga concentrations (geometric mean of 90 $\mu\text{g/g}$ Ga, up to 520 $\mu\text{g/g}$ Ga) are significantly higher than in all other sphalerite species (Table 1 and Fig. 4). In contrast to that, In concentrations of the two Cretaceous sphalerite types are very low (geometric means of 1 $\mu\text{g/g}$ In for Sp IIa and 3 $\mu\text{g/g}$ In for Sp IIb).

Fig. 4 Box-and-whisker plots of the major and trace element geochemistry results of sphalerite from the Freiberg hydrothermal vein deposits, grouped into the four main sphalerite types (Sp Ia, Sp Ib, Sp IIa, and Sp IIb). Zn (wt%) and Fe (wt%) from EPMA analyses. All other elements (Mn, Cd, Cu, Ag, In, Ge, Ga, Co, Sn, As, Sb, Tl, and Pb) are from LA-ICP-MS analyses ($\mu\text{g/g}$) and are plotted logarithmic



Principal component analysis

The results of the PCA applied to the manipulated and log-transformed sphalerite trace element dataset are shown in Fig. 5. The first principal component (PC1), which captures about 50% of the total variance (Fig. 5a), is a measure of Ge, Tl, As, Pb, and Sb concentrations of the sphalerite samples. They all have positive signs and

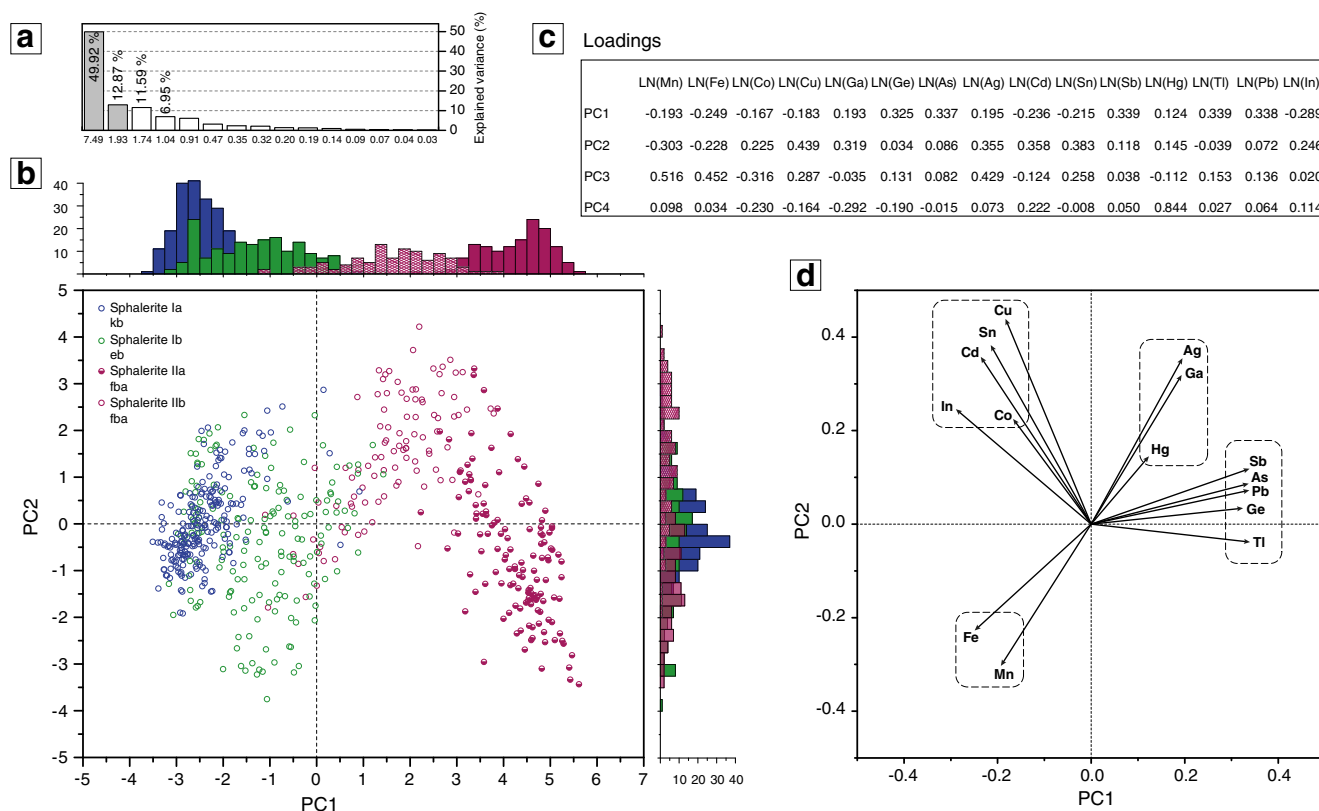


Fig. 5 Principal component analysis (PCA) of the log-transformed LA-ICP-MS dataset of trace element compositions in sphalerite from Freiberg. **a** Scree plot of the eigenvalues of the correlation matrix, explaining the variance. **b** Score plot of all data points, and corresponding histograms of PC1 and PC2 for the trace element

composition of the four sphalerite species. **c** Loadings (i.e., extracted eigenvectors) of the principal components (PC1 to PC4, explaining 81.33% of the variance). **d** Loading plot of the PCA showing the elements (i.e., variables) and framed groups of elements with similar behavior

are positively related (Fig. 5c). PC2 is a measure of the Cu, Sn, Ag, Ga, and Cd concentrations. PC3 is mostly a measure of the Mn and Fe concentrations (Fig. 5c). PC4 is a measure of Hg concentrations. Elements showing similar behavior are highlighted in four clusters in the loading biplot of PC1 vs. PC2 (Fig. 5d). The groupings are almost identical to the main PCs, except that Cu, Sn, Cd, In, and Co, as well as Ag, Ga, and Hg, form a subgroup in the loading plot (Fig. 5d). The angles of the arrows in Fig. 5d are related to the covariance of the element concentrations. If the angle between two arrows is orthogonal, they are uncorrelated. If the angle of two arrows is either close to 0° or 180°, they are positively or negatively correlated. The clearest example here is Mn and Ga, which are negatively correlated. Sphalerite IIb shows very low Mn concentrations but exhibits the highest Ga concentrations (Fig. 4b, d).

Sulfur isotopes

Sulfur isotope compositions of sulfide minerals (sphalerite, galena, pyrite, and arsenopyrite) from 10 samples representing the Permian and the Cretaceous mineralization are summarized

in Fig. 6 and details are given in the Electronic supplementary material (Table ESM 6). Sulfide minerals from the Permian base metal mineralization have $\delta^{34}\text{S}$ values around 0‰ VCDT. Sulfide minerals from the kb veins have $\delta^{34}\text{S}$ values from −1.1 to 0.9‰ VCDT and those from the eb veins yield $\delta^{34}\text{S}$ values between −2.3 and −0.4‰ VCDT. Sulfides of the Cretaceous fba veins have isotopically moderate to extremely negative $\delta^{34}\text{S}$ values that cover a wide range (−30.9 to −5.5‰ VCDT).

Fluid inclusion petrography and microthermometry

Permian (kb and eb veins) mineralization

In total, 204 fluid inclusions of 26 FIAs were measured from the kb and eb vein samples. Neither sphalerite Ib nor carbonate in samples from the eb veins contains fluid inclusions suitable for analysis. The fluid inclusion dataset includes predominantly primary and pseudosecondary fluid inclusions (cf. Table 2 and Table ESM 7). Fluid inclusion studies focused on euhedral Qtz I and Qtz II related to the kb veins (samples Loekb01–07) and selected sulfide samples containing large crystals of sphalerite Ia and

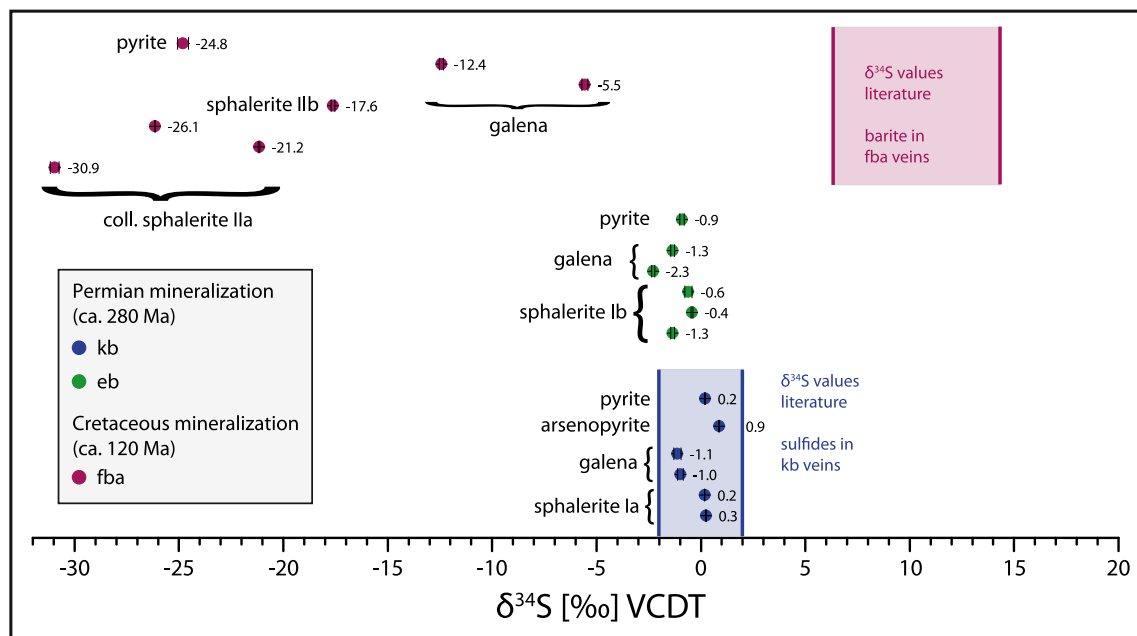


Fig. 6 Sulfur isotope values ($\delta^{34}\text{S}$ [‰] VCDT) of sulfide mineral separates from 10 samples representing the Permian and Cretaceous mineralization. Literature values in blue bar represent the sulfides from

the Permian kb veins (Schrage 1962; Pilot et al. 1970; Haubrich and Tichomirowa 2002; Seifert 2008), and red bar represents literature values for barite from the fba veins (Rösler et al. 1966)

sphalerite Ib (Fig. 7a, d). Comb quartz crystals (Qtz II) exhibit well-developed growth zones, in which the cores and first three epitaxial growth zones are transparent and then change abruptly to milky microcrystalline quartz rims (Fig. 7a–f). Sphalerite Ia formed in interstitials of the quartz crystals, coevally with the milky microcrystalline quartz (Fig. 7a, d). The quartz crystals are overgrown by late ivory-colored bladed carbonates (Fig. 7a).

First visible ice melting occurs at about $-21\text{ }^{\circ}\text{C}$ in all analyzed Permian FIAs. Primary and pseudosecondary aqueous two-phase inclusions are present in clusters in the very first generation of quartz (Qtz I) that formed at the vein selvage (Fig. 7a). FIAs of this generation have homogenization temperatures, T_h (LV $\rightarrow$ L), of 325 to 340 $^{\circ}\text{C}$ and salinities of 5 to 6% eq. $w(\text{NaCl})$ (Table 2). Elongated crystals, most likely sheet silicates (e.g., chlorite), occur irregularly in individual fluid inclusions, indicating accidental trapping. Primary and pseudosecondary FIAs, that are present in the transparent growth zones of Qtz II, have T_h (LV $\rightarrow$ L) of 300 to 325 $^{\circ}\text{C}$ and variable salinities between 0 and 6% eq. $w(\text{NaCl})$ (Table 2). As the FIA types were trapped homogeneously, their T_h (LV $\rightarrow$ L) values represent minimum trapping temperatures. Using Raman spectroscopy, small amounts of CO_2 and CH_4 have been detected in several of the early FIAs, with homogenization temperatures of 300 to 340 $^{\circ}\text{C}$ (Fig. 7b, c). Since these fluid inclusions did not show CO_2 clathrate melting during microthermometric measurements and/or no visible CO_2 was observed at ambient temperatures, the CO_2 concentration has to be less than about 1.5 mol% (Diamond 2001).

Sphalerite Ia from the ore samples hosts aqueous two-phase inclusions, of primary and pseudosecondary origin, in the identical range of homogenization temperatures, T_h (LV $\rightarrow$ L), and salinities as the latter FIA type in quartz (Fig. 7k). In addition to aqueous two-phase fluid inclusions (consistent φ_{vap}) with variable salinities, sphalerite Ia samples and quartz of the kb and eb veins also host plenty of heterogeneously trapped inclusion assemblages of late pseudosecondary and secondary origin: (i) only vapor-rich inclusions (Fig. 7e, f, and j) and (ii) coexisting vapor-rich and two-phase inclusions with a broad range in φ_{vap} (Fig. 7g–i). The minimum homogenization temperatures ($T_{h, \text{min}}$) from two heterogeneously trapped FIAs in quartz of the kb veins are 312 and 310 $^{\circ}\text{C}$, respectively. Final ice melting, T_m (ice), and homogenization, here T_h (LV $\rightarrow$ V), could not be determined in vapor-rich fluid inclusions, because the vapor bubble is too large ($\varphi_{\text{vap}} > 0.95$). FIAs with late pseudosecondary and secondary aqueous two-phase inclusions in the quartz pinnacles of Qtz II and in quartz intergrown with late carbonates have T_h (LV $\rightarrow$ L) of 250 to 290 $^{\circ}\text{C}$ and salinities of 4 to 5% eq. $w(\text{NaCl})$ (Table 2). A pseudosecondary FIA in quartz of the eb veins also shows lower T_h (LV $\rightarrow$ L) around 230 $^{\circ}\text{C}$ and salinities of 1 to 2% eq. $w(\text{NaCl})$ (Table 2). Results of the microthermometric measurements and characteristics of the FIAs are shown in Fig. 9a. The fluid inclusion raw data are included in the Electronic supplementary material (Table ESM 7). Gaseous fluid components, in particular CO_2 (using the same analytical conditions as in Rosso and Bodnar 1995), in these FIAs were not identified by Raman spectroscopy. From this, it follows that the CO_2 concentrations are < 0.13 mol% (Rosso and Bodnar 1995).

Cretaceous fba veins

Fluid inclusion studies were limited to fluorite and sphalerite IIb (Fig. 8a–e). There are no FIAs suitable for analysis in colloform sphalerite IIa, barite, microcrystalline quartz, or chalcedony. In total, 247 fluid inclusions in 25 FIAs of 5 selected samples were analyzed. The size of the fluid inclusions ranges between 10 and 120 μm . First visible melting occurs at about $-52\text{ }^{\circ}\text{C}$ in all analyzed FIAs. Fluid inclusions in fluorite and sphalerite IIb always show T_{m} (hh) before T_{m} (ice) in primary and pseudosecondary inclusions (cf. Fig. 8b). Early FIAs show T_{h} (LV $\rightarrow$ L) around $120\text{ }^{\circ}\text{C}$ and later FIAs, i.e., in outer growth zones of fluorite or sphalerite IIb, have T_{h} at about $105\text{ }^{\circ}\text{C}$ (Table 2 and Fig. 9b). Salinities vary between 17 and 24% eq. $w(\text{NaCl} + \text{CaCl}_2)$ and $\text{Na}/(\text{Na} + \text{Ca})$ ratios are between 0.7 and 0.95 (Fig. 9c).

Discussion

The different sphalerite generations from the major vein types in the Freiberg district show distinctive differences in their trace element composition, although they originate from the same district and within the same host rock environment. While sphalerite Ia from the early Permian kb veins has high concentrations of In, sphalerite IIa and IIb from the Cretaceous fba veins yield elevated Ge and Ga concentrations, respectively (Figs. 4 and 5). Fluid inclusion data and sulfur isotopes allow to identify the specific ore formation mechanisms (Figs. 6 and 9).

Substitution of trace elements in sphalerite

ZnS may incorporate many trace metals, such as In, Ga, Ge, Sn, Cd, Fe, Mn, Co, As, and Tl in solid solution (e.g., Cook et al. 2009). Divalent cations, such as Fe^{2+} , Mn^{2+} , Cd^{2+} , and Co^{2+} , are incorporated in sphalerite in a direct substitution for Zn^{2+} , and sphalerite may contain up to several weight percent of the substituting elements (Barton and Toulmin 1966; Yushkin et al. 1974; Černý and Harris 1978; Balabin and Urusov 1995; Lepetit et al. 2003; Di Benedetto et al. 2005; Wright 2009; Wright and Gale 2010). Whereas tri- and tetravalent cations, like In^{3+} , Ga^{3+} , Ge^{4+} , and Sn^{4+} , can be incorporated in sphalerite in a coupled substitution with Cu^+ and possibly Ag^+ , however, they typically occur in low concentrations in sphalerite (Patrick et al. 1993, 1998; Cook et al. 2009, 2012; Murakami and Ishihara 2013; Belissont et al. 2014). Yet, in some high-temperature Sn-polymetallic granite-related and skarn deposits, sphalerite may contain several weight percent of In in coupled substitution with Cu^+ (e.g., Burke and Kieft 1980; Ohta 1989; Kieft and Damman 1990; Sinclair et al. 2006; Bauer et al. 2017). The

correlation between In and Cu concentrations (Fig. 10a) shows that almost all data points plot along or above the $\text{Cu}/\text{In} = 1:1$ line, suggesting the coupled substitution of $2\text{Zn}^{2+} \leftrightarrow \text{Cu}^+ + \text{In}^{3+}$. The data points that are beyond the 1:1 line toward higher Cu concentrations are probably related to fine-grained chalcopyrite inclusions in sphalerite.

Coupled substitution of Ge^{4+} in colloform sphalerite IIa

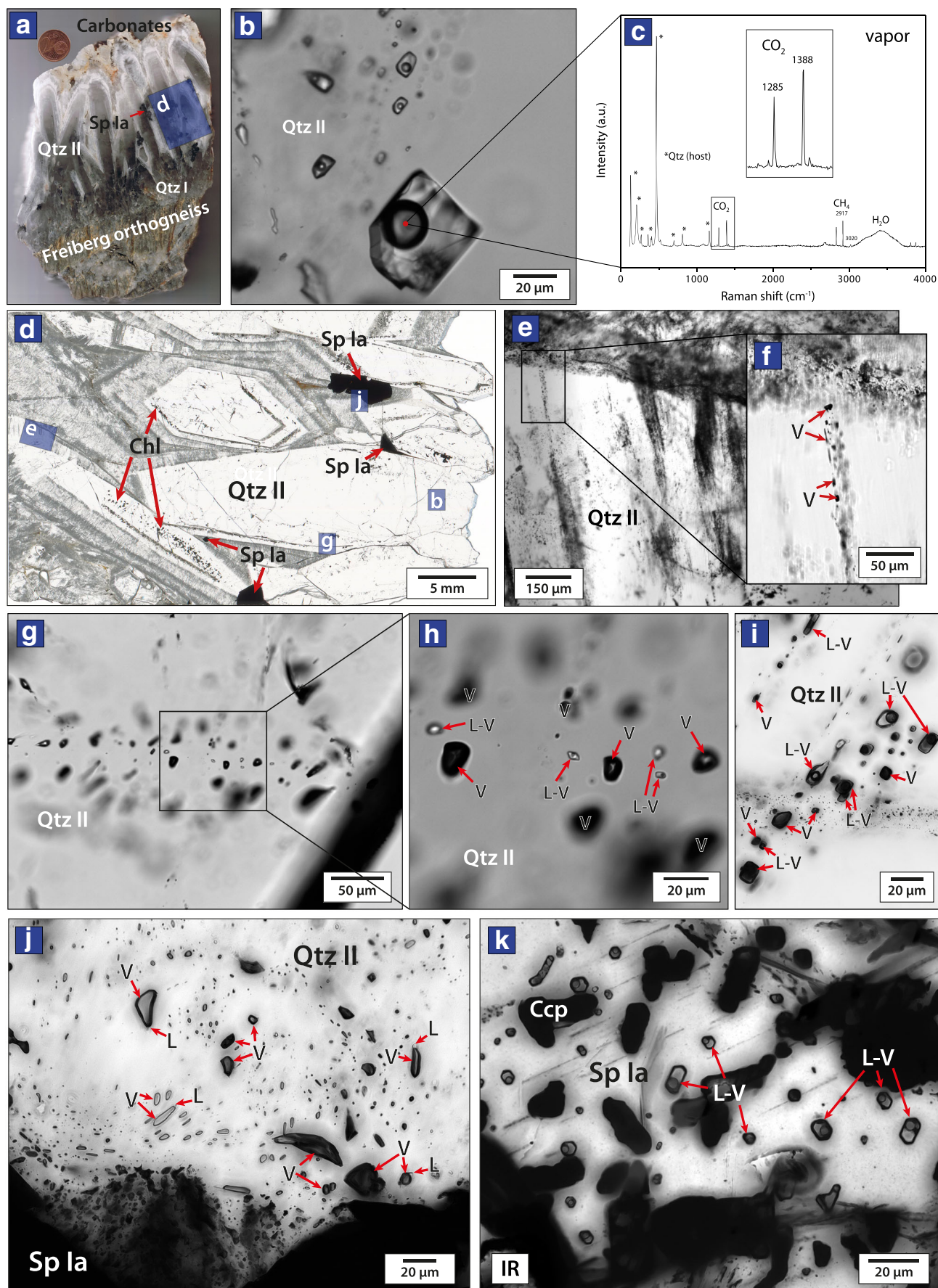
The elements Ge and Cu do not correlate (Fig. 10b, c). In particular, data points of colloform sphalerite IIa are not showing a correlation of Ge and Cu. In contrast, all coarse-grained sphalerite types (Sp Ia, Sp Ib, and Sp IIb) of both mineralization stages follow the expected correlation for Cu vs. In + Ge + Ga (Fig. 10c). Results of synchrotron Ge K-edge μ -XANES studies demonstrated that Ge^{4+} generally occurs in coupled substitution with Cu^+ in sphalerite and there is no evidence to support the presence of Ge^{2+} (Belissont et al. 2016). A bivariate plot of tri- and tetravalent cations vs. monovalent cations (Fig. 10d) indicates that Ge^{4+} incorporation in colloform sphalerite IIb is presumably coupled with Ag^+ . Here, almost all data points of the sphalerite species plot along the $(\text{Cu} + \text{Ag})/(\text{In} + \text{Ge} + \text{Ga} + \text{Sn}) = 1:1$ line or above it (Fig. 10d). An EPMA element distribution map (Fig. 11) illustrates the distribution of Zn, Fe, Cu, Ag, and Ge in colloform sphalerite IIa that is also displayed in Fig. 3j. This element map shows that Ge and Ag are mutually enriched in growth zones of colloform sphalerite IIa (cf. Fig. 11e, f), indicating the coupled substitution of $3\text{Zn}^{2+} \leftrightarrow 2\text{Ag}^+ + \text{Ge}^{4+}$ (cf. Belissont et al. 2014). It is important to note that the zone with the highest Ge and Ag concentrations also coincides with the highest Fe and lowest Cu concentrations in colloform sphalerite IIa (Fig. 11).

Genetic aspects of the Permian mineralization

Fluid pressure estimate of the Permian mineralization

The proximity of the Freiberg hydrothermal veins to Permian intramontaneous volcanosedimentary troughs with its recorded Permian palaeosols (Luthardt and Rößler 2017) as well as findings from zircon and apatite (U–Th)/He and apatite fission track thermochronology (e.g., Ventura and Lisker 2003; Wolff et al. 2015a, b) suggests that the Variscan basement rocks were already eroded during the lower Permian, which means that the stratigraphic level exposed to the surface was already at very shallow crustal depths at that time. Thomas (1982) proposed formation depths of 2–3 km for the Permian mineralization based on boiling FIAs in samples of the kb veins.

To estimate formation pressures/depths, we used the minimum homogenization temperatures ($T_{\text{h, min}} = T_{\text{trap}} = 312$ and



◀ **Fig. 7** Fluid inclusion petrography of the early Permian kb vein samples. **a** Quartz (Qtz) sample Loekb01–07 with contact to gneiss host rock, bladed carbonate overgrows euhedral comb quartz crystals. **b** Homogeneously trapped (consistent $\varphi_{\text{liq}}/\varphi_{\text{vap}}$ ratios) early FIA in transparent Qtz II core. **c** Raman spectrum of vapor bubble in the largest fluid inclusion of **b**; peaks of Qtz host are asterisked; sharp peaks of CO₂ are visible at 1285 and 1388 cm⁻¹ and CH₄ at 2917 cm⁻¹. **d** Double polished thin section sample Loekb02 with transparent Qtz II cores; chlorite (Chl) inclusions in the first visible growth zones; outer milky growth zones are microcrystalline quartz or recrystallized chalcedony; sphalerite Ia (Sp Ia) formed coevally with milky microcrystalline Qtz II. **e** First growth zone with microcrystalline quartz in Qtz II, hosting plenty of primary fluid inclusions, the majority of which are vapor inclusions. **f** Image detail of **e** showing primary trapped vapor inclusions. **g** Pseudosecondary trail of heterogeneously trapped FIA in transparent Qtz II growth zone. **h** Image detail of **g** showing trail of heterogeneously trapped fluid inclusions consisting of dark vapor-rich (V) and two-phase (L-V) inclusions. **i** Heterogeneously trapped FIA consisting of vapor-rich (V) and two-phase (L-V) fluid inclusions in Qtz II. **j** Vapor-rich FIA with $\varphi_{\text{vap}} > 0.95$ in Qtz II next to sphalerite Ia. **k** Infrared photomicrograph of liquid-vapor (L-V) fluid inclusions and chalcopryrite (Ccp) inclusions in sphalerite Ia of sample WSTN1004

310 °C, cf. Fig. 9a) of two heterogeneously trapped FIAs in quartz that directly render a fluid pressure on the boiling curve for a given temperature (after principles of Diamond 2003). The intersection of the given isotherms with the liquid-vapor curve at very low salinities yields a fluid pressure estimate of about 0.1 kbar in the system NaCl-H₂O (Fig. 12; Driesner and Heinrich 2007). These pressure estimates, based on heterogeneous FIAs, correspond to a very shallow formation depth of ~1 km, assuming hydrostatic conditions without taking the low concentrations of CO₂ into account. The p - T diagram in Fig. 12 shows that the salinity, illustrated here with 1 and 6% $w(\text{NaCl})$ solutions (Driesner and Heinrich 2007), has only minor effects on the boiling curve. However, even low concentrations of CO₂, between 0.1 and 1.5 mol% CO₂, which cannot be detected with microthermometry, shift the liquid-vapor curve to higher pressures (e.g., Hedenquist and Henley 1985b; Monecke et al. 2014). The upper limit of CO₂ that we can assume is 1.5 mol%, since higher concentrations would result in the formation of clathrates during freezing and heating in the H₂O-CO₂ system (Diamond 2001). The lower limit of CO₂ that might be present in analyzed fluid inclusions is constrained by the analytical settings for Raman spectroscopy—in our case 0.13 mol% (Rosso and Bodnar 1995). These general considerations in the system H₂O-NaCl-CO₂ could potentially increase the upper pressure limit (200 bars) for the formation of the Permian kb veins to a depth of 2 km (Fig. 12), which roughly coincides with the upper depth limit suggested by Thomas (1982). We presume that this fluid pressure estimate defines an upper hydrostatic depth range of the early Permian mineralization. Additionally, the presence of hydrothermal breccia (e.g., Müller 1901) in many early Permian veins in the ore district of Freiberg indicates fluctuating fluid pressures that may have exceeded the normal hydrostatic

Table 2 Summary of fluid inclusion data

Fluid inclusion type	Host	System	Vein	Fluid inclusion petrography	n FIAs	n fluid inclusions	T_m (ice) range (°C)	T_m (hh) range (°C)	Salinity (range)	T_h (range)	CO ₂ (mol%)
L-V	Qtz I	H ₂ O-NaCl	kb	p and ps	3	21	-2.8 to -3.5	-	4.7 to 5.7	326 to 341 °C	0.1 to 1.5
L-V	Qtz II	H ₂ O-NaCl	kb	p and ps	9	61	-0.3 to -3.8	-	0.5 to 6.2	298 to 324 °C	0.1 to 1.5
L-V	Sp Ia	H ₂ O-NaCl	kb	p, i and ps	7	36	-0.5 to -3.9	-	0.9 to 6.3	285 to 321 °C	n.a.
Coexisting L-V and V	Qtz II	H ₂ O-NaCl	kb	ps	2	43	-0.1 to -1.0	-	0.1 to 1.7	$T_{h, \text{min}} = 310$ °C	n.a.
L-V	Qtz II	H ₂ O-NaCl	kb	ps and s	4	34	-2.5 to -3.2	-	4.2 to 5.3	255 to 294 °C	<0.1
L-V	Qtz	H ₂ O-NaCl	eb	ps and s	2	13	-0.6 to -2.6	-	1.1 to 4.4	225 to 280 °C	<0.1
L-V	Fl	H ₂ O-NaCl-CaCl ₂	fba	p and ps	20	227	-13.2 to -25.1	-14.6 to -28.4	16.9 to 23.8	93 to 140 °C	n.a.
L-V	Sp IIb	H ₂ O-NaCl-CaCl ₂	fba	p and ps	5	20	-14.5 to -19.5	-22.6 to -25.1	17.6 to 21.5	95 to 126 °C	n.a.

Salinities are calculated disregarding possible CO₂ content; H₂O-NaCl system calculated using Steele-MacInnis et al. (2012) and H₂O-NaCl-CaCl₂ system calculated using Steele-MacInnis et al. (2011)
Qtz quartz, Sp sphalerite, Fl fluorite, n.a. not analyzed, p primary, ps pseudosecondary, s secondary, i isolated, hh hydrohalite

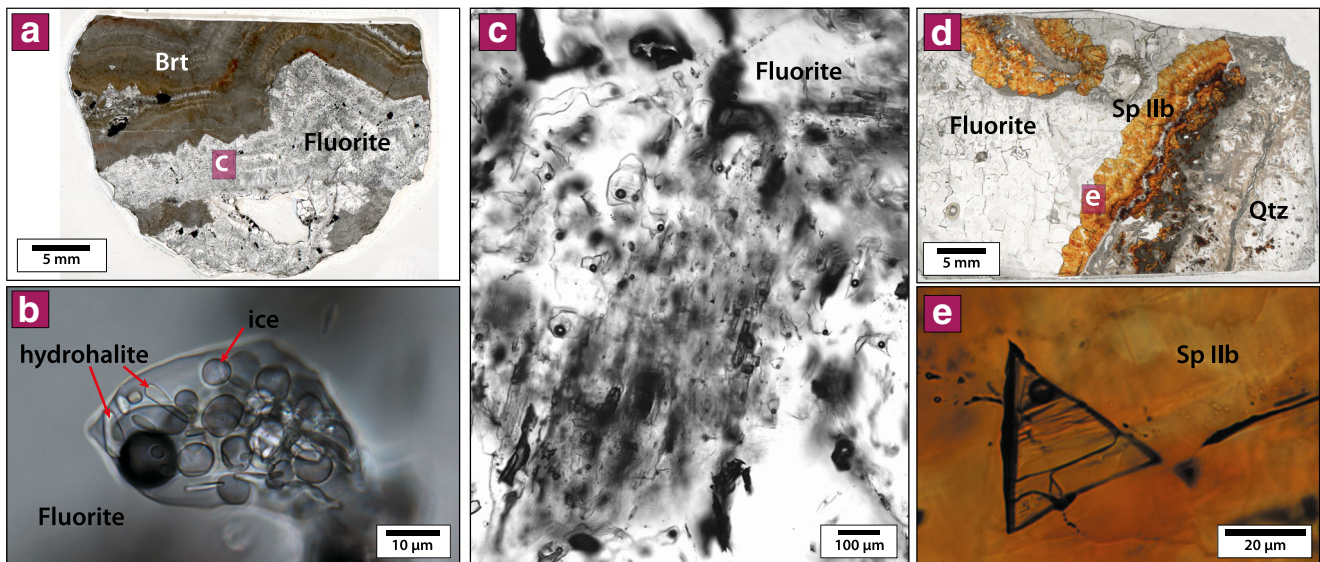


Fig. 8 Fluid inclusion petrography of the Cretaceous fba type samples. **a** Double polished thin section of sample fba17b with fluorite, barite (Brt), and chalcedony. **b** Large fluid inclusion in fluorite with blue ice and yellow hydrohalite at $-23.8\text{ }^{\circ}\text{C}$. **c** Plenty of two-phase fluid inclusions in different sizes with consistent $\varphi_{\text{liq}}/\varphi_{\text{vap}}$ ratios in fluorite;

black regions are large fluid inclusions that were opened by polishing. **d** Double polished thin section of sample fba12 with sphalerite IIb, quartz, and fluorite. **e** Photomicrograph of two-phase fluid inclusion in sphalerite IIb

pressure temporarily and that hydrodynamic conditions dominated (cf. Hedenquist and Henley 1985a).

Fluid boiling in the Permian mineralization

Several details in the formation of the early Permian hydrothermal veins are brought together to constrain the crucial ore-forming processes. The parental ore-forming fluid is low to medium saline (0 and 6% eq. $w[\text{NaCl}]$) at temperatures around 340 to 300 $^{\circ}\text{C}$ (Fig. 9a). Fluid *boiling* is evident in coexisting vapor-rich and liquid-vapor FIAs (Fig. 7e–j). This has been observed before by Thomas (1982) in ore samples of the kb veins. Fluid boiling might be attributed to rock failure, which caused a drop of the fluid pressure below vapor saturation in the hydrothermal fluid system. Characteristic quartz textures (cf. outer milky microcrystalline quartz rims in Fig. 7a, d, and Sander and Black 1988; Dong et al. 1995) and in particular yellow cathodoluminescence (CL) in hydrothermal quartz from the kb veins were interpreted to indicate fast crystallization (Götze et al. 2015) probably resulting from these boiling processes. Paragenetically late bladed carbonates (cf. Fig. 7a) are considered characteristic for epithermal fluid systems, that form as a consequence of fluid boiling and exsolution of CO_2 (e.g., Catchpole et al. 2011; Moncada et al. 2017). This is supported by Raman spectroscopy, which shows that low amounts of CO_2 (and CH_4) are present in the first FIAs with homogenization temperatures of 300 to 340 $^{\circ}\text{C}$, and the last generations of fluid inclusions (T_h from 290 to 220 $^{\circ}\text{C}$) associated with carbonate

precipitation have no detectable CO_2 . Similar observations about the gas content (CO_2 and CH_4) in fluid inclusions of the Permian hydrothermal veins have been made by Klemm (1994).

Ore precipitation mechanism

The initial metal sulfide precipitation in the early Permian mineralization was likely triggered by water-rock interaction and boiling of the ascending ore-forming fluid, which led to subsequent drastic cooling of the ore fluid (Fig. 9). Chloride complexes that transported the base metals, e.g., Zn, Pb, Cu, Ag, Sn, and In, are dissociated by cooling (caused by boiling), and in consequence, metal sulfides (like ZnS) precipitate and H^+ is released from H_2S . The mineral assemblage of quartz, pyrite, and kaolinite group minerals in the veinward side of the early Permian wall rock alteration represents an advanced argillic alteration (cf. Fig. 2). Since dickite and nacrite polymorphs are apparently present in the wall rocks of the early Permian vein mineralization (most prevalent in the eb veins) and pyrophyllite is obviously missing in the alteration assemblage (Ross and Kerr 1930; Rösler and Kühne 1970), the advanced argillic alteration probably did not form until the mineralizing fluid was colder than 290 $^{\circ}\text{C}$. At temperatures above 290 $^{\circ}\text{C}$, pressures < 1 kbar, and an assumed H_2O activity > 0.9 , quartz and the kaolinite group minerals react to pyrophyllite (Evans and Guggenheim 1988). As the factors controlling the formation of dickite and

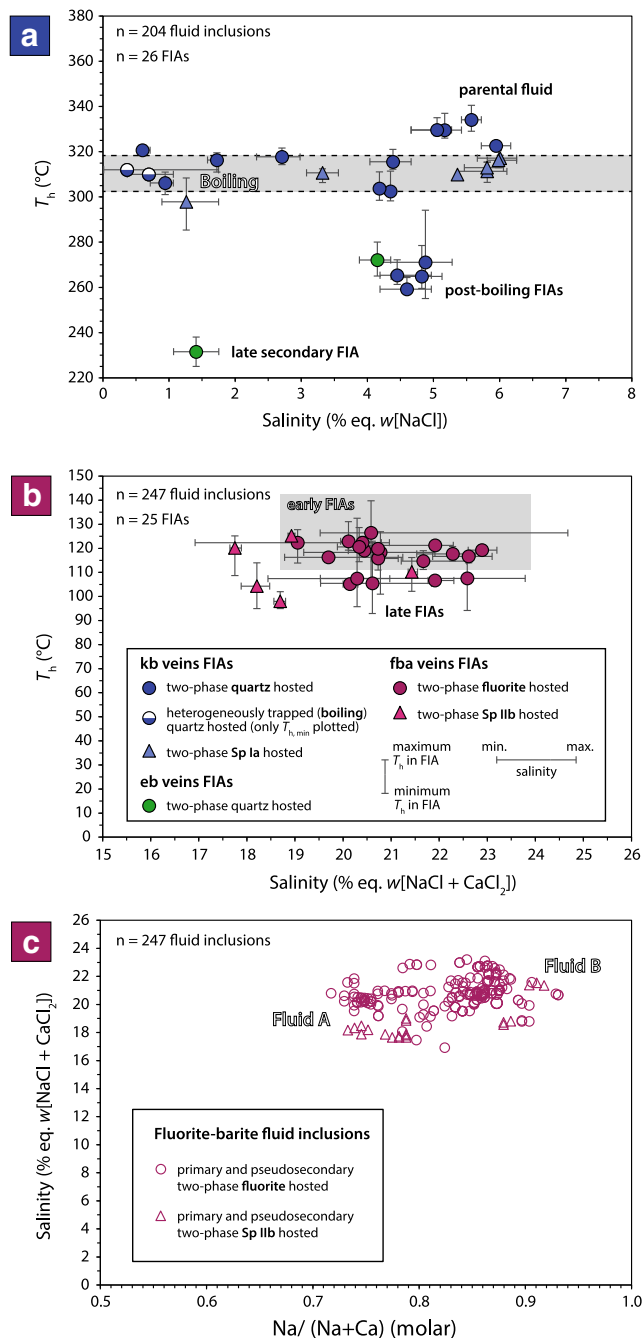


Fig. 9 Results of microthermometric measurements of various fluid inclusion assemblages (FIAs). **a** Homogenization temperature (T_h in °C) vs. salinity (in % eq. $w[\text{NaCl}]$) of FIAs in the early Permian kb and eb veins; figure legend in **b**. **b** Homogenization temperature (T_h in °C) vs. salinity (in % eq. $w[\text{NaCl} + \text{CaCl}_2]$) of FIAs in the Cretaceous fba veins. **c** Salinity (in % eq. $w[\text{NaCl} + \text{CaCl}_2]$) vs. $\text{Na}/(\text{Na} + \text{Ca})$ ratio of all analyzed fluid inclusions of the Cretaceous fba veins

nacrite are not fully understood, their presence in the wall rock alteration alone cannot be used to determine minimum formation temperatures (Wilson 2013). In a greater distance to the base metal veins, a sericite-chlorite to propylitic alteration is prevalent (Rösler and

Kühne 1970). Fluid inclusion data from this study indicates that carbonates and the abundant Ag- and Ag-Sb-sulfides of the eb veins formed at lower temperatures (less than 300 °C) than the kb veins (Fig. 9), which is in perfect agreement to the fluid inclusion data reported in Burisch et al. (2018a).

Genetic aspects of the Cretaceous fluorite-barite mineralization

The variation in fluid salinities (17 to 24% eq. $w[\text{NaCl} + \text{CaCl}_2]$) and $\text{Na}/(\text{Na} + \text{Ca})$ ratios (0.7 to 0.95) at invariably low temperatures of c. 120 °C (Fig. 9) and the large scatter in $\delta^{34}\text{S}$ (Fig. 6) indicate that the Cretaceous fluorite-barite (fba) veins formed by mixing of two highly saline brines. The data is in perfect agreement to previously published unconformity-related formation models of central and western European fluorite-barite-Pb-Zn-Ag-mineralization (e.g., Trinkler et al. 2005; Staude et al. 2009; Boiron et al. 2010; Fusswinkel et al. 2013; Burisch et al. 2016; Walter et al. 2016; Sośnicka and Lüders 2018). Based on the fluid inclusion data, mineralogy, and stable isotopic compositions, we propose that the fba veins of the Freiberg district analogously formed due to mixing of a deep-seated basement brine and a sedimentary fluid (Fusswinkel et al. 2013; Burisch et al. 2016; Walter et al. 2016). Episodic mixing events (Bons and van Milligen 2001; Burisch et al. 2016, 2018b) during simultaneous upward movement (Bons et al. 2014) of chemically contrasting fluids are an efficient ore-forming process for Pb-Zn-Ag-fluorite-barite veins. Such types of Pb-Zn-Ag-fluorite-barite-veins occur widespread in central Europe and are genetically related to the extensional stress regime, which was present during the opening of the northern Atlantic (Mitchell and Halliday 1976; Boiron et al. 2010; Ostendorf et al. 2018). However, the formation depths of the Freiberg fba veins are difficult to constrain, since the thickness of the eroded sedimentary cover is not well known. The oldest preserved Mesozoic sedimentary rocks are Upper Cretaceous conglomerates and fluvial sandstones in the greater region around Freiberg (e.g., Janetschke and Wilmsen 2014) that were probably deposited more than 20 Ma later (Cenomanian) than the supposed formation age (c. 120 Ma) of the fba veins. Zircon and apatite (U-Th)/He thermochronology suggest that the thickness of the eroded Mesozoic sedimentary cover and basement in the region around Freiberg was about 2 to 3 km (Wolff et al. 2015a). Assuming a geothermal gradient of 30 to 40 °C for sedimentary basins, the average homogenization temperature of 120 °C implies a formation depth between 3 and 4 km (hydrostatic). However, it has to be taken into account that the fluid after mixing is most likely not in thermal equilibrium with the host rock (Burisch et al.

Fig. 10 Bivariate plots of **a** Cu vs. In, **b** Cu vs. Ge, **c** Cu vs. In + Ge + Ga, and **d** Cu + Ag vs. In + Ge + Ga + Sn

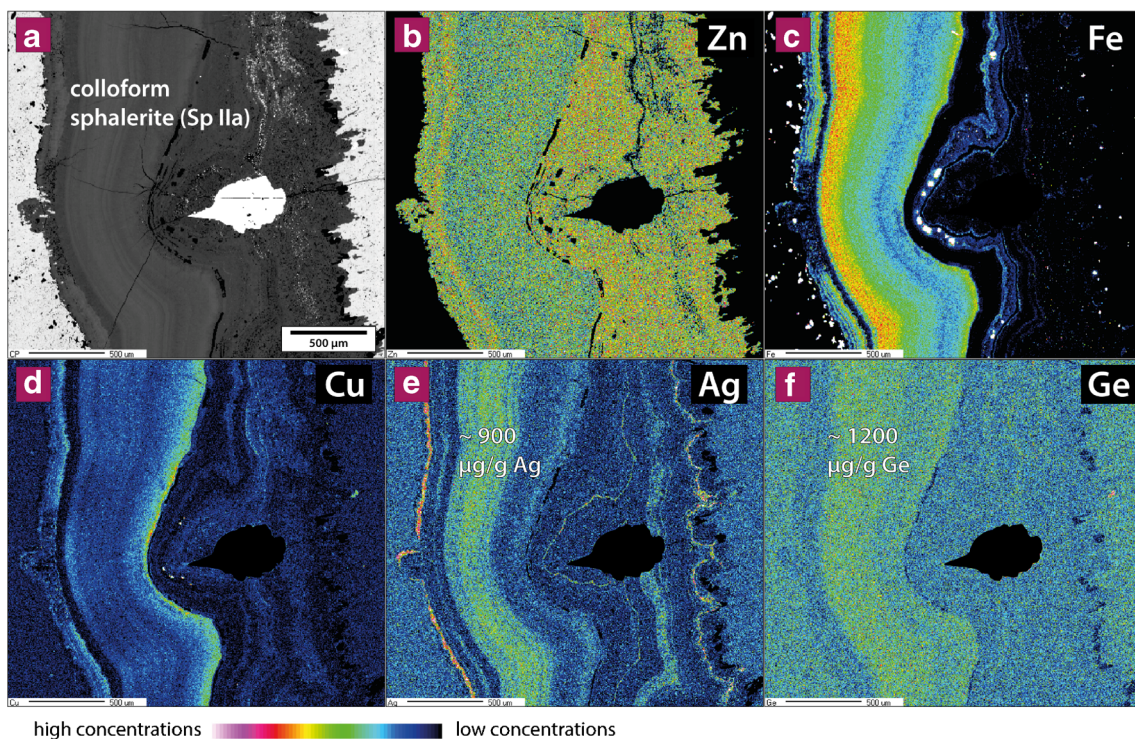
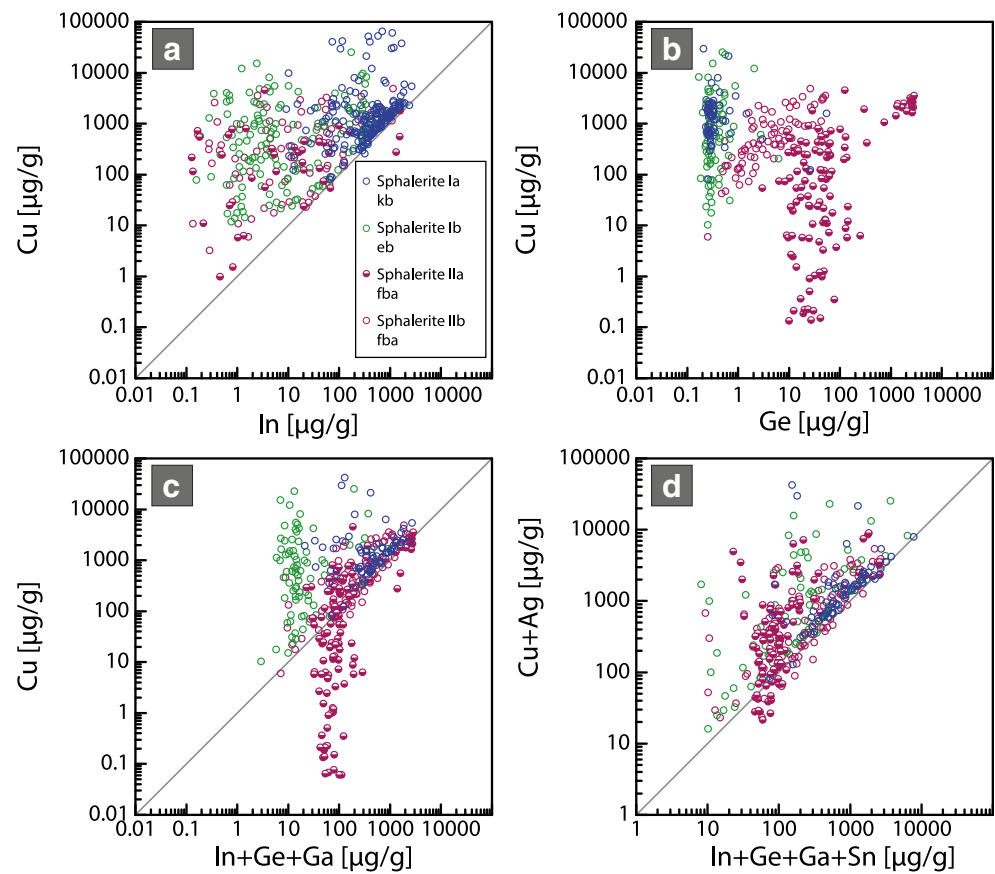


Fig. 11 EPMA element distribution maps of Cretaceous colloform sphalerite IIa (sample fba18); resolution 1 pixel $\cong$ 4 μ m. **a** Backscatter

electron image. **b** Zn EDS signal. **c** Fe K α X-ray signal. **d** Cu K α X-ray signal. **e** Ag L α X-ray signal. **f** Ge L α X-ray signal

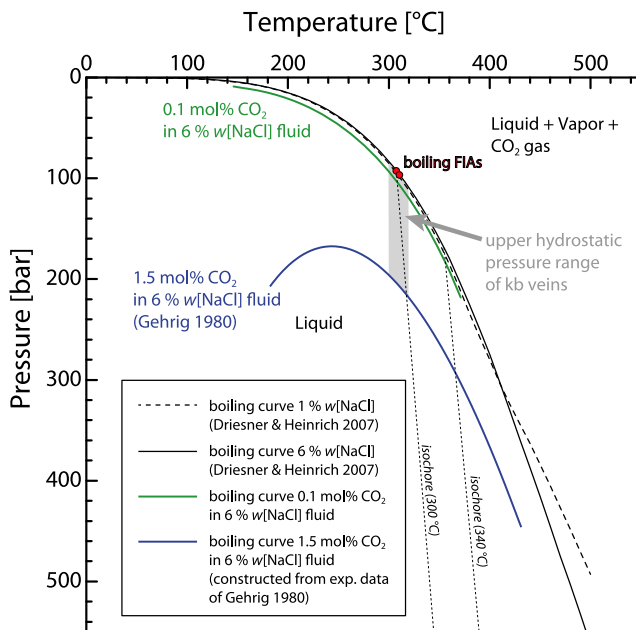


Fig. 12 *p*-*T* conditions for kb veins; pressure estimates based on heterogeneous FIAs; boiling curve for the H₂O-NaCl system (for 1% w(NaCl) and 6% w(NaCl) solutions) was determined using software from Driesner and Heinrich (2007); boiling curve with 0.1 mol% CO₂ in 6% w(NaCl) solution was calculated using Henry's law and principles outlined in Hedenquist and Henley (1985b) and references therein; boiling curve with 1.5 mol% CO₂ in 6% w(NaCl) solution was constructed using experimental data by Gehrig (1980)

2017), since the mixture is dominated by a hotter ascending basement brine. Therefore, the formation depth is most likely 0.5 to 1 km shallower than the homogenization temperatures imply. Due to the high salinities, pressure corrections for the homogenization temperatures (T_h) result in only minor temperature deviations (< 10 °C) and are therefore not necessary.

Fluid sources

Permian (kb and eb veins) mineralization

Sulfur isotope data for both Permian vein samples (kb and eb) presented here show values that are comparable to literature data (cf. Fig. 6). The small range of $\delta^{34}\text{S}$ values of the Permian Zn-Pb mineralization between -2 and $+2\text{‰}$ VCDT strongly indicate a magmatic-hydrothermal source. Potential fluid and heat sources in the Freiberg mining district are most likely related to early Permian felsic magmatism that occurred in the Erzgebirge and the foreland basins (Ostendorf et al. 2018). Subvolcanic and volcanic rocks in the Flöha fault zone and the Erzgebirge (Chemnitz) Basin in close proximity to Freiberg have ages between 270 and 310 Ma, with an age maximum between 280 and 290 Ma (e.g., Seifert 2008; Luthardt et al. 2018). This is also supported by

the observation of boiling in fluid inclusions, the fluid chemistry with its low salinity (characteristic for magmatic-hydrothermal vein systems), and the trace element geochemistry of Permian Sp Ia and to some extent of Sp Ib, which have higher concentrations of In, Co, Sn, and Mn. Elevated concentrations in sphalerite of these elements seem to be typical for magmatic-hydrothermal fluid systems (cf. Cook et al. 2009).

Cretaceous fba veins

The large scatter in $\delta^{34}\text{S}$ between -5.5 and -30.9‰ VCDT of sulfide mineral separates of the Cretaceous fluorite-barite mineralization and $\delta^{34}\text{S}$ values of associated barite of $+6$ to $+14\text{‰}$ VCDT (cf. Fig. 6 and Rösler et al. 1966) indicate fluid mixing of two highly saline brines and partial reduction of marine sulfate. Possible sulfur sources for the fba veins include (i) sulfate from Upper Permian evaporates (sabkha deposits representing a coastal facies of the “Zechstein”) that are present in the Erzgebirge Basin (Schneider et al. 2012) and/or (ii) marine sulfate of Cretaceous origin. The sulfur isotope record of marine sulfate in the Permian is between $+20$ and $+25\text{‰}$ VCDT, and Cretaceous marine $\delta^{34}\text{S}$ is lower between $+15$ and $+20\text{‰}$ VCDT (Kampschulte and Strauss 2004). The $\delta^{34}\text{S}$ systematics in the Cretaceous fba mineralization are similar to other unconformity-related fluorite-barite veins in central Europe (e.g., Schwinn et al. 2006) and indicate mixing of a basement-derived brine with a sedimentary brine that contains marine sulfate (e.g., Boiron et al. 2010; Fusswinkel et al. 2013; Burisch et al. 2016; Walter et al. 2016).

Temperature dependency of trace element geochemistry in sphalerite

Several authors suggested that the trace element concentrations in sphalerite reflect the origin from certain hydrothermal deposits or indicate the temperature of formation (e.g., Oftedal 1941; Schroll 1954, 1955; Cook et al. 2009; Ye et al. 2011; Belissont et al. 2014; Frenzel et al. 2016a). The recently proposed GGIMFis geothermometer (Frenzel et al. 2016a) was applied to the LA-ICP-MS dataset of this work (cf. Fig. 5). The average of the calculated temperatures of formation (T_{calc} with GGIMFis) and uncertainties from the four different sphalerite types (Sp I to IV) are plotted together with fluid data, that is partially pressure corrected (Fig. 13). The upper pressure limit (0.1 to 0.2 kbar) is given by the boiling FIAs of the kb veins (Figs. 7 and 9a). Consistently, the boiling FIAs do not have to be pressure corrected. Yet, as the trapping pressure cannot be lower than the homogenization pressure, the FIAs of the early Permian mineralization with higher

sphalerite Ib and associated carbonates make it difficult to constrain possible reasons for the lower In concentrations of Sp Ib compared to Sp Ia. Thus, the very low In concentrations in Cretaceous Sp Ila and Sp I Ib (geometric means of 1 µg/g In for Sp Ila and 3 µg/g In for Sp I Ib), that precipitated in a completely different fluid system (high salinities, low temperatures, unconformity-related), at least imply that the In concentrations in the fba fluids were relatively low. Unconformity-related basement fluids are typically highly concentrated in base metals (Fusswinkel et al. 2013; Burisch et al. 2016), which are believed to originate from water-rock interaction of these highly saline fluids with crystalline basement rocks. Although these fluids are capable of transporting large amounts of base metals (due to their high chlorinity), the fluid conditions seem not to be appropriate to mobilize or transport In. Alternatively, it might be a reason of the availability of In, since the contribution of a magmatic fluid component is unlikely for the Cretaceous mineralization. Conversely, the high Ge concentrations of Sp Ila (up to 1700 µg/g Ge) and elevated Ga concentrations of Sp I Ib (up to 520 µg/g Ga) suggest that the Cretaceous fba fluids were substantially enriched in Ga and Ge (in addition to Zn, Pb, and Ag).

There is little but sufficient data on the behavior of Ga and Ge in natural hydrothermal fluid systems. It is known from experimental complexation studies in aqueous geochemistry (up to 300 °C) that Ga and Ge form stable complexes with hydroxide, fluoride, and sulfate, while chloride complexes are negligible (Wood and Samson 2006). This is in strong contrast to In that seems to form unilaterally strong chloride complexes (Seward et al. 2000; Ashworth and Frisch 2017). The presence of abundant fluorite and barite in the Cretaceous fba mineralization suggests that fluoride and possibly sulfate complexation may have also been important in the mobilization and transport of Ga and Ge in the Cretaceous basement brines. The findings in this study are also supported by generalities in geochemistry: Ga and Ge are expected to be preferentially concentrated in major rock-forming silicates, due to their geochemical similarities to Al and possibly Si (e.g., Shaw 1957; Bernstein 1985; Höll et al. 2007). This promotes their availability to large-scale hydrothermal leaching in crustal basement rocks in general (Burisch et al. 2016). In comparison to Ga and Ge, In (together with Sn) has a stronger mixed-affinity character in geochemical processes and In is therefore particularly enriched in highly fractionated felsic Sn-specialized granitic intrusion systems (Jenner 2017). In conclusion, enrichment of In, Ga, and Ge in hydrothermal mineral deposits is likely a result of both source rock availability and effective complexation chemistry.

Conclusions

This study demonstrates that trace element concentrations of critical metals (In, Ge, and Ga) in sphalerite related to different mineralization events within the Freiberg ore district are systematically different. The four paragenetic and textural generations of sphalerite are related to two distinct ore-forming events. Sphalerite of the first mineralization stage (Sp Ia and Ib) formed in response to boiling and cooling of a low-salinity, most likely magmatic-hydrothermal fluid at temperatures at or above 300 °C during the early Permian. Sphalerite of the second mineralization stage (Sp Ila and Sp I Ib) formed in the Cretaceous by mixing of high-salinity fluids with contrasting Na/(Na + Ca) ratios of 0.7 and 0.95 at temperatures around 120 °C. Only the first generation of sphalerite (Sp Ia) is significantly enriched in In—in comparison to the other three sphalerite generations. In contrast, colloform sphalerite (Sp Ila) of Cretaceous age has the highest concentrations of Ge, whereas Ga concentrations are elevated in coarse-grained sphalerite I Ib.

The results of this study demonstrate that the trace element geochemistry of sphalerite (e.g., systematics in relative concentrations of Ga, Ge, In, Mn, and Fe in the GGIMFis geothermometer) and associated fluid temperatures correlate remarkably well (cf. Frenzel et al. 2016a). The observed trace element trends in the sphalerite types support findings that In (together with Sn, Co, and Mn) is specifically enriched in high-temperature sulfide deposits that are related to magmatic-hydrothermal fluid systems, while Ge and Ga are more concentrated in low-temperature deposits that are considered to not contain magmatic fluid components (cf. Cook et al. 2009; Belissont et al. 2014). The study illustrates that such differences in sphalerite mineral chemistry are indeed controlled by physicochemical conditions during sphalerite precipitation (e.g., temperature, Frenzel et al. 2016a) but may also be influenced by fluid source, mechanisms of sphalerite precipitation, and the effective complexation chemistry of In, Ge, and Ga in the ore-forming fluid systems.

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